

Detecting and Reducing Model Dependence in Causal Inference

Gary King¹

Institute for Quantitative Social Science
Harvard University

Peking University, 1/26/2017

¹GaryKing.org

Readings, Part 1: Model Dependence

Readings, Part 1: Model Dependence

- King, Gary and Langche Zeng. “The Dangers of Extreme Counterfactuals,” *Political Analysis*, 14, 2, (2007): 131-159.

Readings, Part 1: Model Dependence

- King, Gary and Langche Zeng. “The Dangers of Extreme Counterfactuals,” *Political Analysis*, 14, 2, (2007): 131-159.
- King, Gary and Langche Zeng. “When Can History be Our Guide? The Pitfalls of Counterfactual Inference,” *International Studies Quarterly*, 2006, 51 (March, 2007): 183–210.

Readings, Part 1: Model Dependence

- King, Gary and Langche Zeng. “The Dangers of Extreme Counterfactuals,” *Political Analysis*, 14, 2, (2007): 131-159.
- King, Gary and Langche Zeng. “When Can History be Our Guide? The Pitfalls of Counterfactual Inference,” *International Studies Quarterly*, 2006, 51 (March, 2007): 183–210.
- Related Software: WhatIf, MatchIt, Zelig, CEM

Readings, Part 1: Model Dependence

- King, Gary and Langche Zeng. “The Dangers of Extreme Counterfactuals,” *Political Analysis*, 14, 2, (2007): 131-159.
- King, Gary and Langche Zeng. “When Can History be Our Guide? The Pitfalls of Counterfactual Inference,” *International Studies Quarterly*, 2006, 51 (March, 2007): 183–210.
- Related Software: WhatIf, MatchIt, Zelig, CEM

<http://j.mp/causalinference>

Counterfactuals

Counterfactuals

- Three types:

Counterfactuals

- Three types:
 1. **Forecasts** What will the mortality rate be in 2025?

Counterfactuals

- Three types:
 1. **Forecasts** What will the mortality rate be in 2025?
 2. **Whatif Questions** What would have happened if the U.S. had not invaded Iraq?

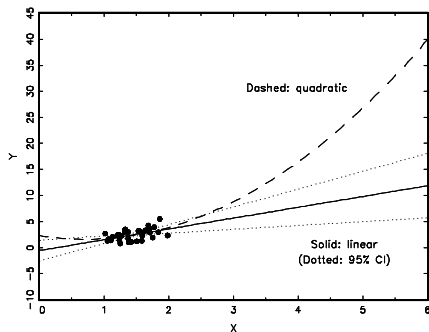
Counterfactuals

- Three types:
 1. **Forecasts** What will the mortality rate be in 2025?
 2. **Whatif Questions** What would have happened if the U.S. had not invaded Iraq?
 3. **Causal Effects** What is the causal effect of the Iraq war on World GDP? (a factual minus a counterfactual)

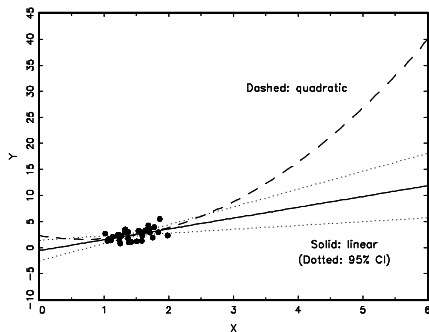
Counterfactuals

- Three types:
 1. **Forecasts** What will the mortality rate be in 2025?
 2. **Whatif Questions** What would have happened if the U.S. had not invaded Iraq?
 3. **Causal Effects** What is the causal effect of the Iraq war on World GDP? (a factual minus a counterfactual)
- Counterfactuals are part of most social science research

Which model would you choose? (Both fit the data well.)

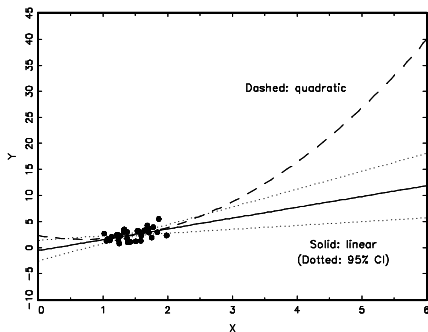


Which model would you choose? (Both fit the data well.)



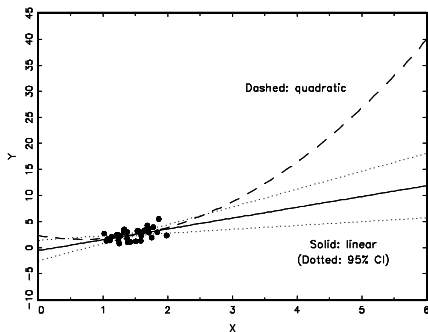
- Compare prediction at $x = 1.5$ to prediction at $x = 5$

Which model would you choose? (Both fit the data well.)



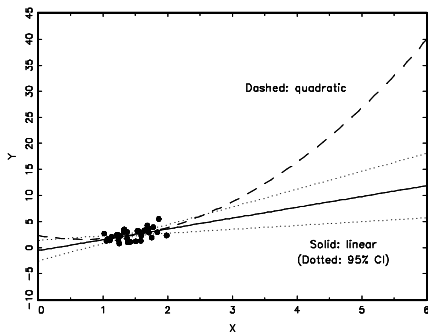
- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model?

Which model would you choose? (Both fit the data well.)



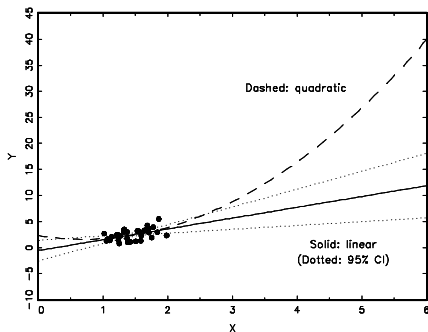
- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model? R^2 ?

Which model would you choose? (Both fit the data well.)



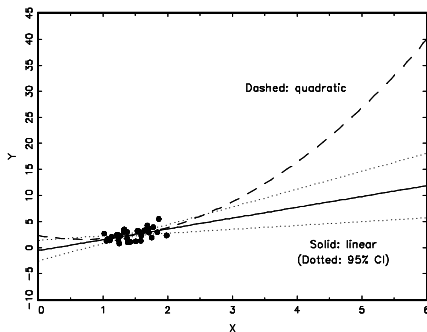
- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model? R^2 ? Some “test”?

Which model would you choose? (Both fit the data well.)



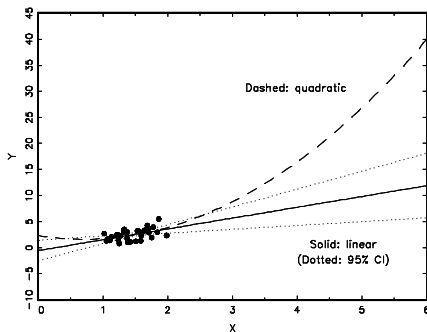
- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model? R^2 ? Some “test”? “Theory”?

Which model would you choose? (Both fit the data well.)



- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model? R^2 ? Some “test”? “Theory”?
- The bottom line: answers to some questions don’t exist in the data. We show how to determine which ones.

Which model would you choose? (Both fit the data well.)



- Compare prediction at $x = 1.5$ to prediction at $x = 5$
- How do you choose a model? R^2 ? Some “test”? “Theory”?
- The bottom line: answers to some questions don’t exist in the data. We show how to determine which ones.
- Same for what if questions, predictions, and causal inferences

Model Dependence Proof

Model Dependence Proof

Model Free Inference

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Assumptions (Model-Based Inference)

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Assumptions (Model-Based Inference)

1. Definition: model dependence at x is the difference between predicted outcomes for any two models that fit about equally well.

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Assumptions (Model-Based Inference)

1. Definition: model dependence at x is the difference between predicted outcomes for any two models that fit about equally well.
2. The functional form follows strong continuity (think smoothness, although it is less restrictive)

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Assumptions (Model-Based Inference)

1. Definition: model dependence at x is the difference between predicted outcomes for any two models that fit about equally well.
2. The functional form follows strong continuity (think smoothness, although it is less restrictive)

Result

Model Dependence Proof

Model Free Inference

To estimate $E(Y|X = x)$ at x , average many observed Y with value x

Assumptions (Model-Based Inference)

1. Definition: model dependence at x is the difference between predicted outcomes for any two models that fit about equally well.
2. The functional form follows strong continuity (think smoothness, although it is less restrictive)

Result

The maximum degree of model dependence: a function of the distance from the counterfactual to the data

A Simple Measure of Distance from The Data

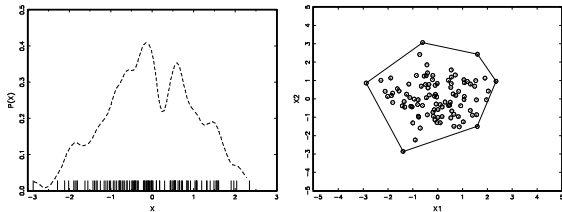


Figure: The Convex Hull

A Simple Measure of Distance from The Data

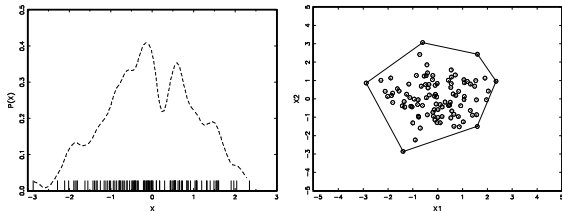


Figure: The Convex Hull

- **Interpolation:** Inside the convex hull

A Simple Measure of Distance from The Data

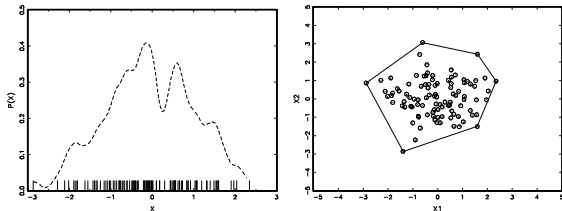


Figure: The Convex Hull

- **Interpolation:** Inside the convex hull
- **Extrapolation:** Outside the convex hull

A Simple Measure of Distance from The Data

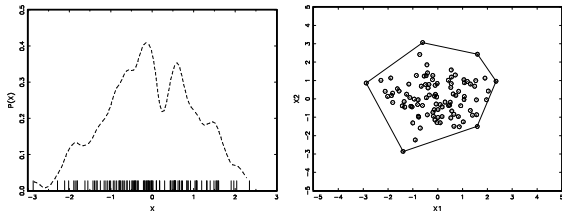


Figure: The Convex Hull

- **Interpolation:** Inside the convex hull
- **Extrapolation:** Outside the convex hull
- Works mathematically for any number of X variables

A Simple Measure of Distance from The Data

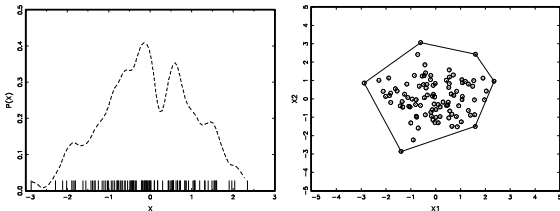


Figure: The Convex Hull

- **Interpolation:** Inside the convex hull
- **Extrapolation:** Outside the convex hull
- Works mathematically for any number of X variables
- Software to determine whether a point is in the hull (which is all we need) without calculating the hull (which would take forever), so its fast; see GaryKing.org/whatif

Model Dependence Example

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model
- **The question:** How *model dependent* are the results?

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model
- **The question:** How *model dependent* are the results?
- **Percent of counterfactuals in the convex hull:**

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model
- **The question:** How *model dependent* are the results?
- **Percent of counterfactuals in the convex hull:** 0%

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model
- **The question:** How *model dependent* are the results?
- **Percent of counterfactuals in the convex hull:** 0%
- $\rightsquigarrow$ **without estimating any models, we know:** inferences will be model dependent

Model Dependence Example

Replication of Doyle and Sambanis, APSR 2000

(From: King and Zeng, 2007)

- **Data:** 124 Post-World War II civil wars
- **Dependent var:** peacebuilding success
- **Treatment:** multilateral UN peacekeeping intervention (0/1)
- **Control vars:** war type, severity, duration; development status, . . .
- **Counterfactual question:** Switch UN intervention for each war
- **Data analysis:** Logit model
- **The question:** How *model dependent* are the results?
- **Percent of counterfactuals in the convex hull:** 0%
- $\rightsquigarrow$ without estimating any models, we know: inferences will be model dependent
- **For illustration:** let's find an example. . . .

Two Logit Models, Apparently Similar Results

Two Logit Models, Apparently Similar Results

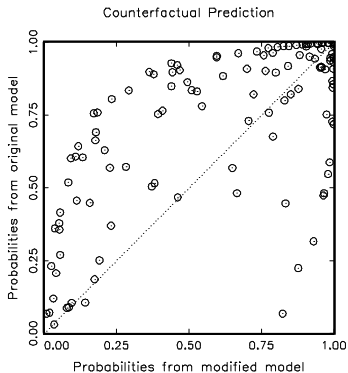
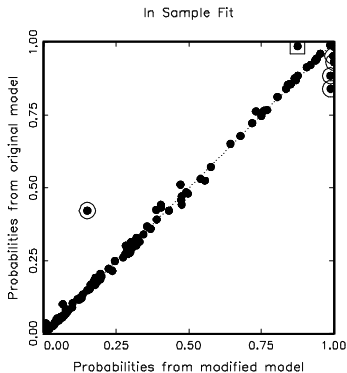
Effect of Multilateral UN Intervention on Peacebuilding Success

Two Logit Models, Apparently Similar Results

Effect of Multilateral UN Intervention on Peacebuilding Success

Variables	Original “Interactive” Model			Modified Model		
	Coeff	SE	P-val	Coeff	SE	P-val
Wartype	−1.742	.609	.004	−1.666	.606	.006
Logdead	−.445	.126	.000	−.437	.125	.000
Wardur	.006	.006	.258	.006	.006	.342
Factnum	−1.259	.703	.073	−1.045	.899	.245
Factnum2	.062	.065	.346	.032	.104	.756
Trnsfcap	.004	.002	.010	.004	.002	.017
Develop	.001	.000	.065	.001	.000	.068
Exp	−6.016	3.071	.050	−6.215	3.065	.043
Decade	−.299	.169	.077	−0.284	.169	.093
Treaty	2.124	.821	.010	2.126	.802	.008
UNOP4	3.135	1.091	.004	.262	1.392	.851
Wardur*UNOP4	—	—	—	.037	.011	.001
Constant	8.609	2.157	0.000	7.978	2.350	.000
N	122			122		
Log-likelihood	−45.649			−44.902		
Pseudo R^2	.423			.433		

Model Dependence: Same Fit, Different Predictions



Readings, Part 2: Matching

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?

⇒ “Causal Inference Without Balance Checking: Coarsened Exact Matching” (PA, 2011. Stefano Iacus, Gary King, and Giuseppe Porro)

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?
 $\rightsquigarrow$ “Causal Inference Without Balance Checking:
 Coarsened Exact Matching” (PA, 2011. Stefano
 Iacus, Gary King, and Giuseppe Porro)
2. The most popular method (propensity score matching, used in
 76,900 articles!) sounds magical:

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?
~→ “Causal Inference Without Balance Checking: Coarsened Exact Matching” (PA, 2011. Stefano Iacus, Gary King, and Giuseppe Porro)
2. The most popular method (propensity score matching, used in 76,900 articles!) sounds magical:
~→ “Why Propensity Scores Should Not Be Used for Matching” (Gary King, Richard Nielsen)

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?
~→ “Causal Inference Without Balance Checking: Coarsened Exact Matching” (PA, 2011. Stefano Iacus, Gary King, and Giuseppe Porro)
2. The most popular method (propensity score matching, used in 76,900 articles!) sounds magical:
~→ “Why Propensity Scores Should Not Be Used for Matching” (Gary King, Richard Nielsen)
3. Matching methods optimize either imbalance ($\approx$ bias) or # units pruned ($\approx$ variance); users need both simultaneously’:

Readings, Part 2: Matching

1. Do powerful methods have to be complicated?
~> “Causal Inference Without Balance Checking: Coarsened Exact Matching” (PA, 2011. Stefano Iacus, Gary King, and Giuseppe Porro)
2. The most popular method (propensity score matching, used in 76,900 articles!) sounds magical:
~> “Why Propensity Scores Should Not Be Used for Matching” (Gary King, Richard Nielsen)
3. Matching methods optimize either imbalance ($\approx$ bias) or # units pruned ($\approx$ variance); users need both simultaneously':
~> “The Balance-Sample Size Frontier in Matching Methods for Causal Inference” (*AJPS*; Gary King, Christopher Lucas and Richard Nielsen)

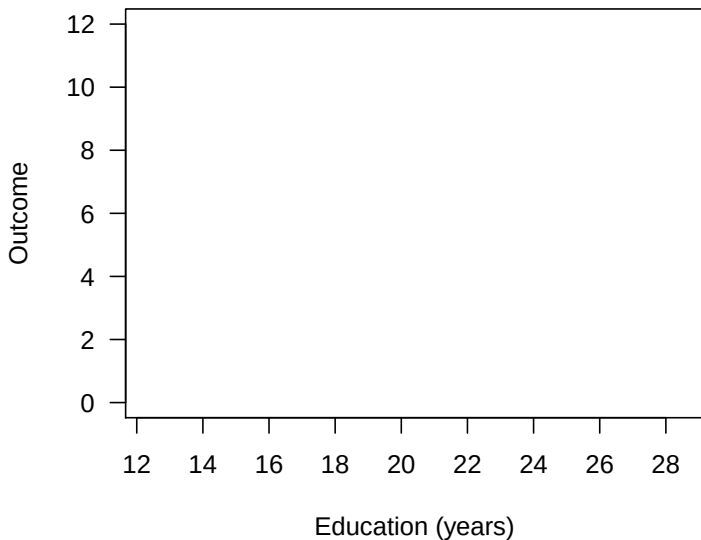
Matching to Reduce Model Dependence

Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)

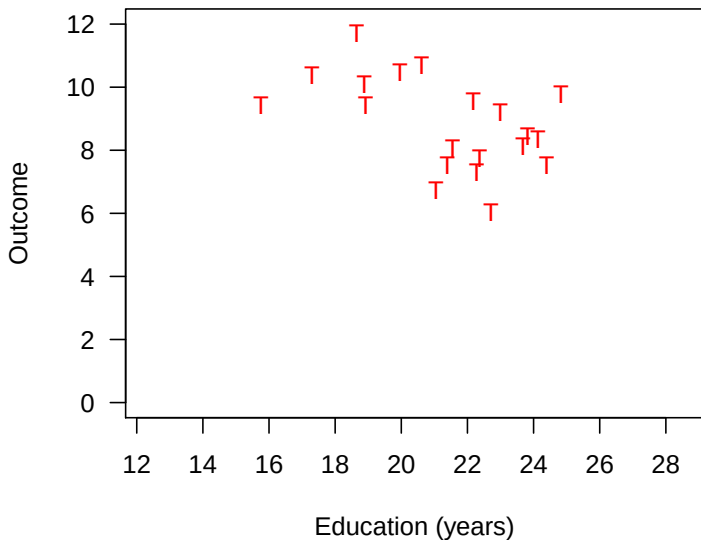
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



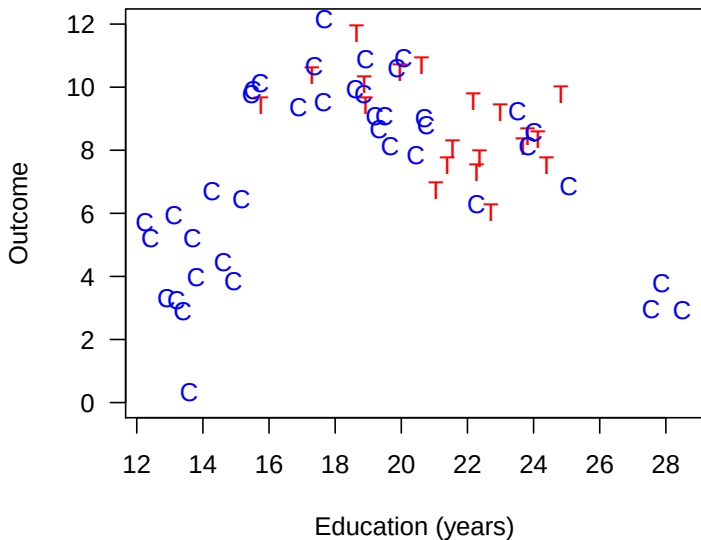
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



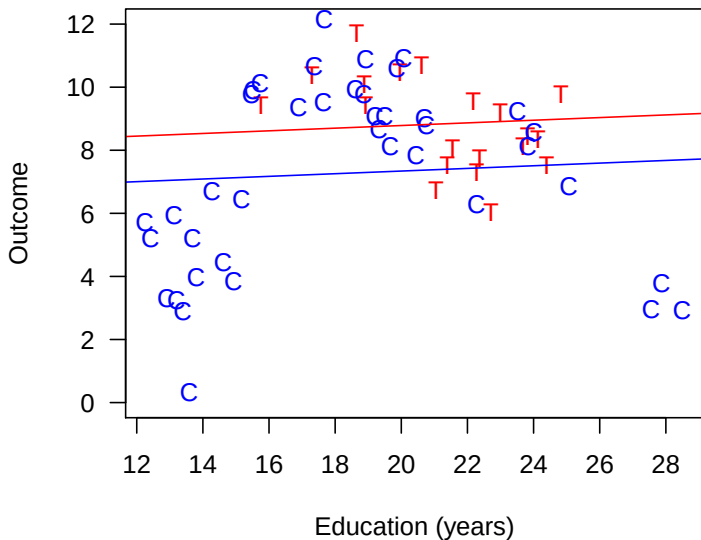
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



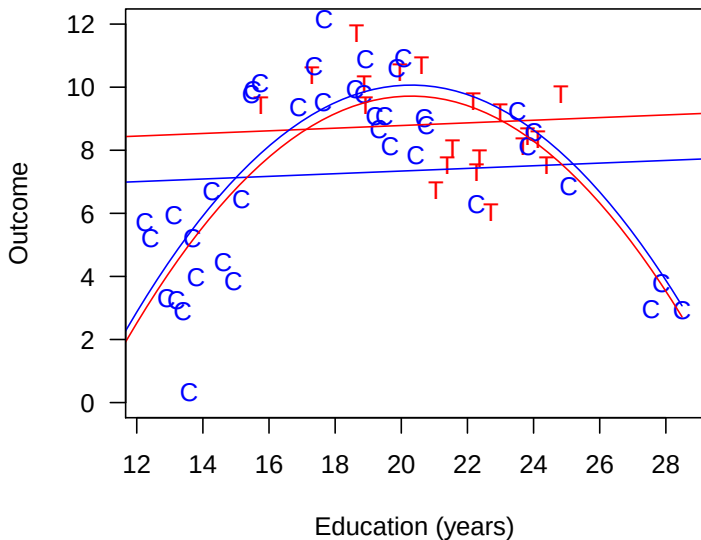
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



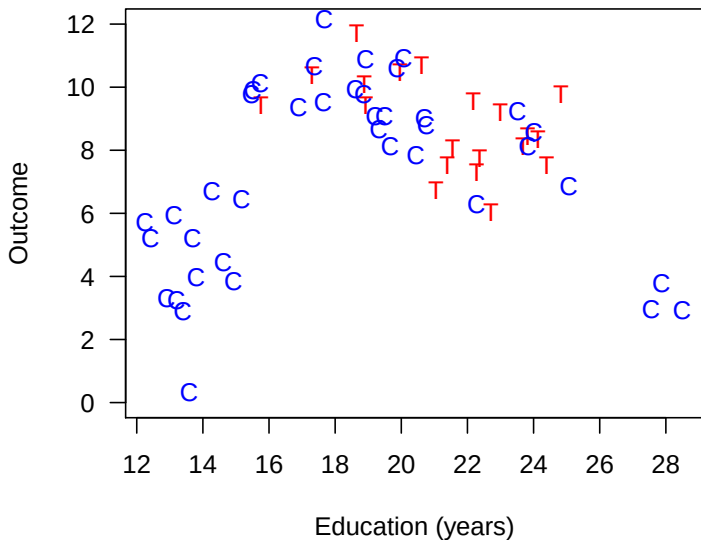
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



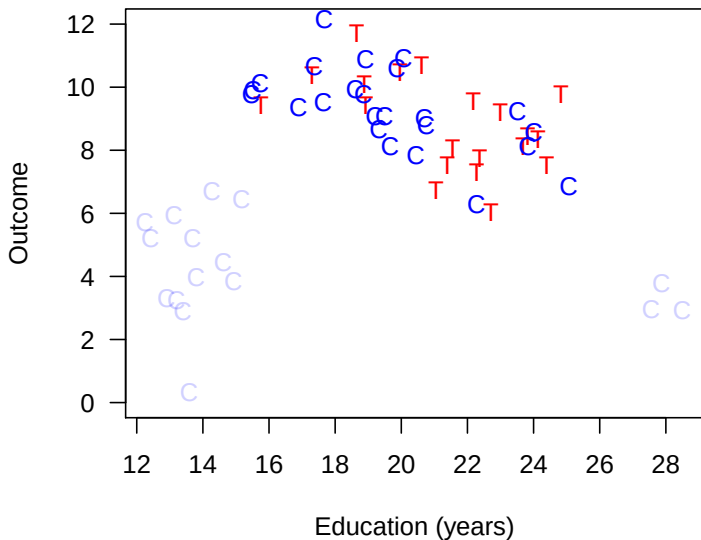
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



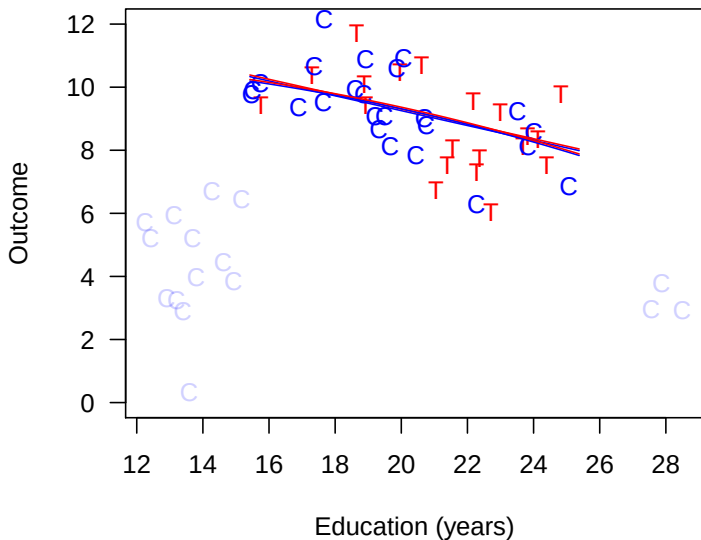
Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



Matching to Reduce Model Dependence

(Ho, Imai, King, Stuart, 2007: fig.1, *Political Analysis*)



The Problems Matching Solves

The Problems Matching Solves

Without Matching:

The Problems Matching Solves

Without Matching:

Imbalance

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments
 - Choosing based on “plausibility” is probably worse_[eff]

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments
 - Choosing based on “plausibility” is probably worse_[eff]
- conscientious effort doesn't avoid biases (Banaji 2013)_[acc]

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments
 - Choosing based on “plausibility” is probably worse_[eff]
- conscientious effort doesn't avoid biases (Banaji 2013)_[acc]
- People do not have easy access to their own mental processes or feedback to avoid the problem (Wilson and Brekke 1994)_[exprt]

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments
 - Choosing based on “plausibility” is probably worse_[eff]
- conscientious effort doesn't avoid biases (Banaji 2013)_[acc]
- People do not have easy access to their own mental processes or feedback to avoid the problem (Wilson and Brekke 1994)_[exprt]
- Experts overestimate their ability to control personal biases more than nonexperts, and more prominent experts are the most overconfident (Tetlock 2005)_[tch]

The Problems Matching Solves

Without Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

- Qualitative choice from unbiased estimates = biased estimator
 - e.g., Choosing from *results* of 50 randomized experiments
 - Choosing based on “plausibility” is probably worse_[eff]
- conscientious effort doesn't avoid biases (Banaji 2013)_[acc]
- People do not have easy access to their own mental processes or feedback to avoid the problem (Wilson and Brekke 1994)_[exprt]
- Experts overestimate their ability to control personal biases more than nonexperts, and more prominent experts are the most overconfident (Tetlock 2005)_[tch]
- “Teaching psychology is mostly a waste of time” (Kahneman 2011)

The Problems Matching Solves

~~Without~~ Matching:

Imbalance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

The Problems Matching Solves

~~Without~~ Matching:

~~Im~~balance $\rightsquigarrow$ Model Dependence $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

The Problems Matching Solves

~~Without~~ Matching:

~~Imbalance~~ $\rightsquigarrow$ ~~Model Dependence~~ $\rightsquigarrow$ Researcher discretion $\rightsquigarrow$ Bias

The Problems Matching Solves

Without Matching:

~~Imbalance~~ $\rightsquigarrow$ ~~Model Dependence~~ $\rightsquigarrow$ ~~Researcher discretion~~ $\rightsquigarrow$ Bias

The Problems Matching Solves

Without Matching:

~~Imbalance~~ $\rightsquigarrow$ ~~Model Dependence~~ $\rightsquigarrow$ ~~Researcher discretion~~ $\rightsquigarrow$ ~~Bias~~

The Problems Matching Solves

Without Matching:

~~Imbalance~~ $\rightsquigarrow$ ~~Model Dependence~~ $\rightsquigarrow$ ~~Researcher discretion~~ $\rightsquigarrow$ ~~Bias~~

A central project of statistics: Automating away human discretion

What's Matching?

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$TE_i = Y_i(1) - Y_i(0)$$

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i(1) - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control
- Quantities of Interest:

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control
- Quantities of Interest:
 1. SATT: Sample Average Treatment effect on the Treated:

$$SATT = \text{Mean}_{i \in \{T_i=1\}} (TE_i)$$

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control
- Quantities of Interest:
 1. SATT: Sample Average Treatment effect on the Treated:

$$SATT = \text{Mean}_{i \in \{T_i=1\}} (TE_i)$$

2. FSATT: Feasible SATT (prune badly matched treateds too)

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} \text{TE}_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control
- Quantities of Interest:
 1. SATT: Sample Average Treatment effect on the Treated:

$$\text{SATT} = \text{Mean}_{i \in \{T_i=1\}} (\text{TE}_i)$$

2. FSATT: Feasible SATT (prune badly matched treateds too)
- **Big convenience:** Follow preprocessing with whatever statistical method you'd have used without matching

What's Matching?

- Y_i dep var, T_i (1=treated, 0=control), X_i confounders
- Treatment Effect for treated observation i :

$$\begin{aligned} TE_i &= Y_i - Y_i(0) \\ &= \text{observed} - \text{unobserved} \end{aligned}$$

- Estimate $Y_i(0)$ with Y_j with a matched ($X_i \approx X_j$) control
- Quantities of Interest:
 1. SATT: Sample Average Treatment effect on the Treated:

$$SATT = \text{Mean}_{i \in \{T_i=1\}} (TE_i)$$

2. FSATT: Feasible SATT (prune badly matched treateds too)
- **Big convenience:** Follow preprocessing with whatever statistical method you'd have used without matching
 - **Pruning nonmatches makes control vars matter less:** reduces imbalance, model dependence, researcher discretion, & bias

Empirical Illustration: Carpenter, AJPS, 2002

Empirical Illustration: Carpenter, AJPS, 2002

- Hypothesis: Democratic senate majorities slow FDA drug approval time

Empirical Illustration: Carpenter, AJPS, 2002

- Hypothesis: Democratic senate majorities slow FDA drug approval time
- $n = 408$ new drugs (262 approved, 146 pending).

Empirical Illustration: Carpenter, AJPS, 2002

- Hypothesis: Democratic senate majorities slow FDA drug approval time
- $n = 408$ new drugs (262 approved, 146 pending).
- lognormal survival model.

Empirical Illustration: Carpenter, AJPS, 2002

- Hypothesis: Democratic senate majorities slow FDA drug approval time
- $n = 408$ new drugs (262 approved, 146 pending).
- lognormal survival model.
- 18 control variables (clinical factors, firm characteristics, media variables, etc.)

Evaluating Reduction in Model Dependence

Evaluating Reduction in Model Dependence

- Focus on the causal effect of a Democratic majority in the Senate (identified by Carpenter as not robust).

Evaluating Reduction in Model Dependence

- Focus on the causal effect of a Democratic majority in the Senate (identified by Carpenter as not robust).
- Match: prune 49 units (2 treated, 17 control units).

Evaluating Reduction in Model Dependence

- Focus on the causal effect of a Democratic majority in the Senate (identified by Carpenter as not robust).
- Match: prune 49 units (2 treated, 17 control units).
- run 262,143 possible specifications and calculates SATT for each.

Evaluating Reduction in Model Dependence

- Focus on the causal effect of a Democratic majority in the Senate (identified by Carpenter as not robust).
- Match: prune 49 units (2 treated, 17 control units).
- run 262,143 possible specifications and calculates SATT for each.
- Look at *variability* in SATT estimate across specifications.

Evaluating Reduction in Model Dependence

- Focus on the causal effect of a Democratic majority in the Senate (identified by Carpenter as not robust).
- Match: prune 49 units (2 treated, 17 control units).
- run 262,143 possible specifications and calculates SATT for each.
- Look at *variability* in SATT estimate across specifications.
- (Normal applications would only use one or a few specifications.)

Reducing Model Dependence

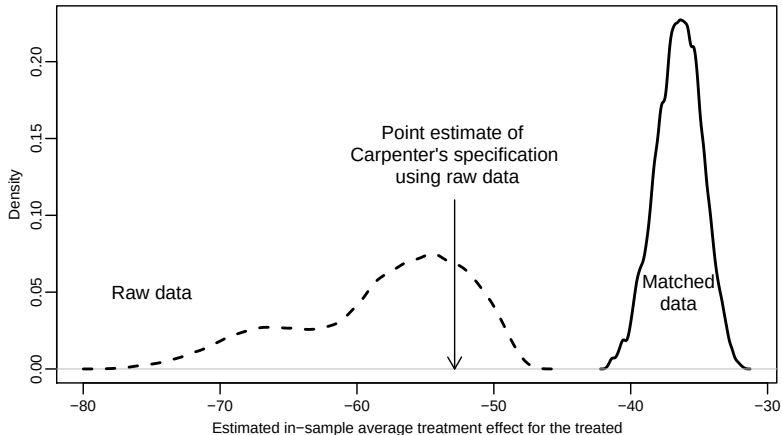


Figure: SATT Histogram: Effect of Democratic Senate majority on FDA drug approval time, across 262,143 specifications.

Another Example: Jeffrey Koch, AJPS, 2002

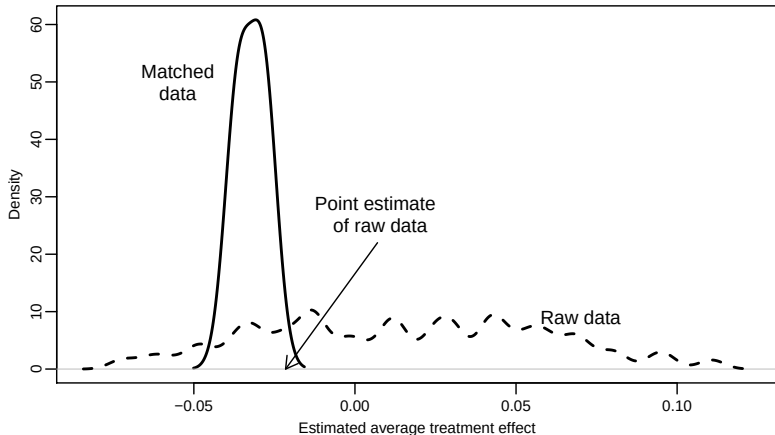


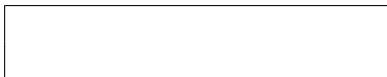
Figure: SATT Histogram: Effect of being a highly visible female Republican candidate across 63 possible specifications with the Koch

Matching: Finding Hidden Randomized Experiments

Matching: Finding Hidden Randomized Experiments

Matching: Finding Hidden Randomized Experiments

Types of Experiments



Matching: Finding Hidden Randomized Experiments

Types of Experiments

*Complete
Randomization*



Matching: Finding Hidden Randomized Experiments

Types of Experiments

<i>Complete Randomization</i>	<i>Fully Blocked</i>

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>		
<i>Unobserved</i>		

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	
<i>Unobserved</i>		

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	
<i>Unobserved</i>	On average	

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

$\rightsquigarrow$ *Fully blocked dominates complete randomization*

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

$\rightsquigarrow$ *Fully blocked* dominates *complete randomization* for:

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

$\rightsquigarrow$ *Fully blocked* dominates *complete randomization* for:
imbalance,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency, bias,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency, bias, research costs,

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency, bias, research costs, robustness.

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for:
imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Goal of Each Matching Method (in Observational Data)

Matching: Finding Hidden Randomized Experiments

Types of Experiments

	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for: imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Goal of Each Matching Method (in Observational Data)

- PSM: *complete randomization*

Matching: Finding Hidden Randomized Experiments

Types of Experiments

	<i>Complete</i>	<i>Fully</i>
Balance		
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for: imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Goal of Each Matching Method (in Observational Data)

- PSM: *complete randomization*
- Other methods: *fully blocked*

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for: imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Goal of Each Matching Method (in Observational Data)

- PSM: *complete randomization*
- Other methods: *fully blocked*
- **Other matching methods dominate PSM**

Matching: Finding Hidden Randomized Experiments

Types of Experiments

Balance	<i>Complete</i>	<i>Fully</i>
Covariates:	<i>Randomization</i>	<i>Blocked</i>
<i>Observed</i>	On average	Exact
<i>Unobserved</i>	On average	On average

⇒ *Fully blocked* dominates *complete randomization* for: imbalance, model dependence, power, efficiency, bias, research costs, robustness. E.g., Imai, King, Nall 2009: SEs 600% smaller!

Goal of Each Matching Method (in Observational Data)

- PSM: *complete randomization*
- Other methods: *fully blocked*
- **Other matching methods dominate PSM** (wait, it gets worse)

Method 1: Mahalanobis Distance Matching

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. **Preprocess** (Matching)
2. **Estimation** Difference in means or a model

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)' S^{-1} (X_c - X_t)}$

2. Estimation Difference in means or a model

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)' S^{-1} (X_c - X_t)}$
- (Mahalanobis is for methodologists; in applications, use Euclidean!)

2. Estimation Difference in means or a model

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)'S^{-1}(X_c - X_t)}$
- (Mahalanobis is for methodologists; in applications, use Euclidean!)
- Match each treated unit to the nearest control unit

2. Estimation Difference in means or a model

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)'S^{-1}(X_c - X_t)}$
- (Mahalanobis is for methodologists; in applications, use Euclidean!)
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused

2. Estimation Difference in means or a model

Method 1: Mahalanobis Distance Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)' S^{-1} (X_c - X_t)}$
- (Mahalanobis is for methodologists; in applications, use Euclidean!)
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused
- Prune matches if $\text{Distance} > \text{caliper}$

2. Estimation Difference in means or a model

Method 1: Mahalanobis Distance Matching

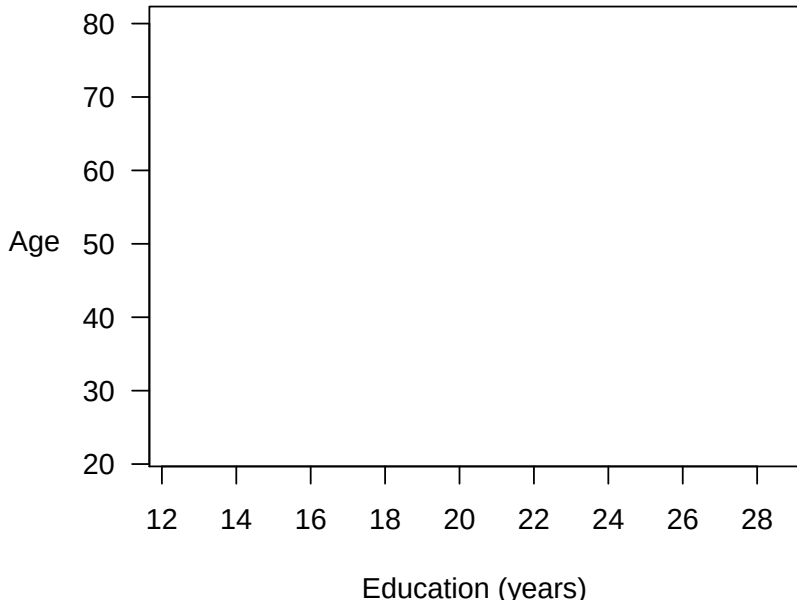
(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

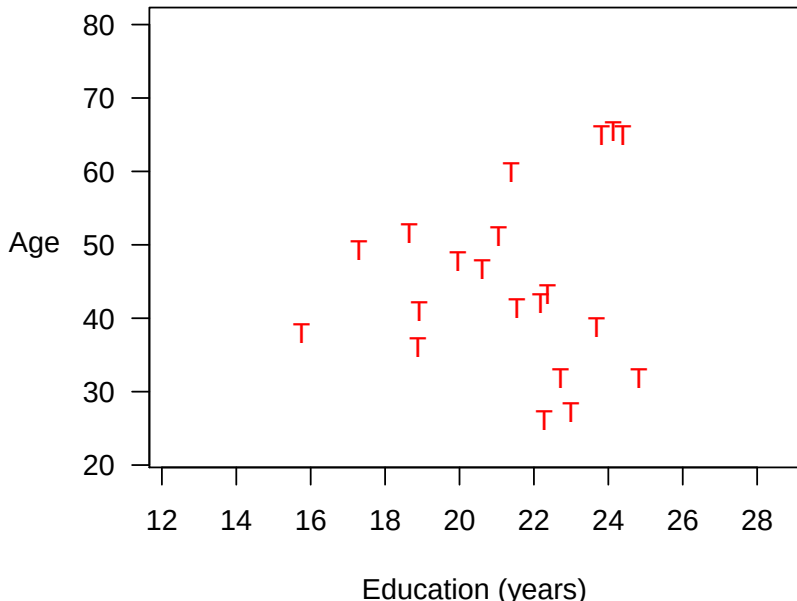
- $\text{Distance}(X_c, X_t) = \sqrt{(X_c - X_t)'S^{-1}(X_c - X_t)}$
- (Mahalanobis is for methodologists; in applications, use Euclidean!)
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused
- Prune matches if $\text{Distance} > \text{caliper}$
- (Many adjustments available to this basic method)

2. Estimation Difference in means or a model

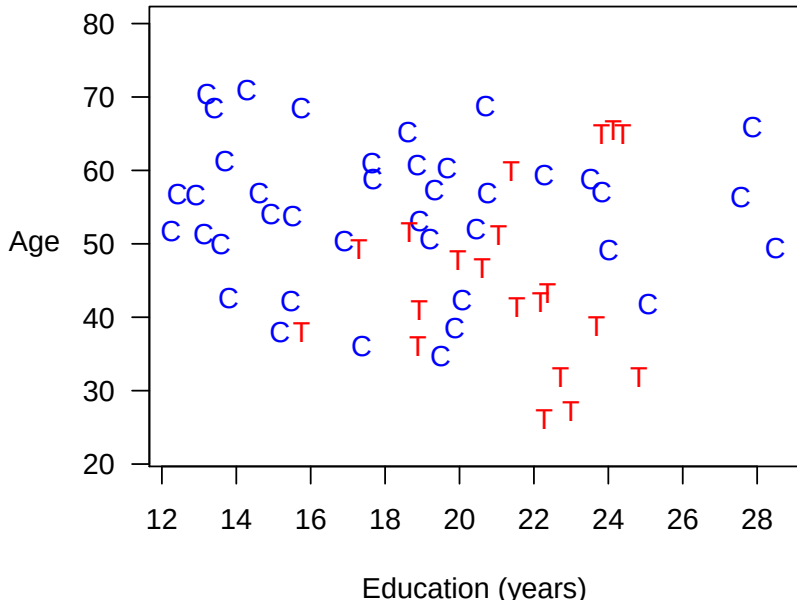
Mahalanobis Distance Matching



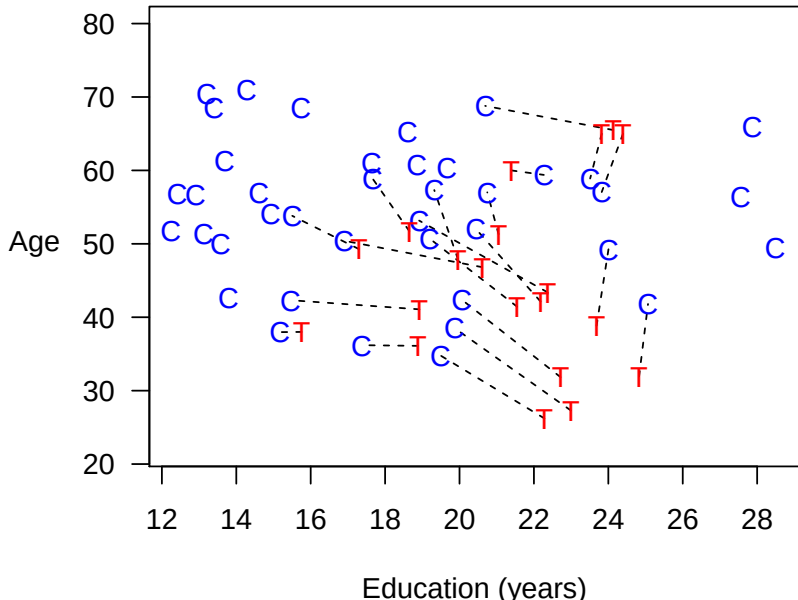
Mahalanobis Distance Matching



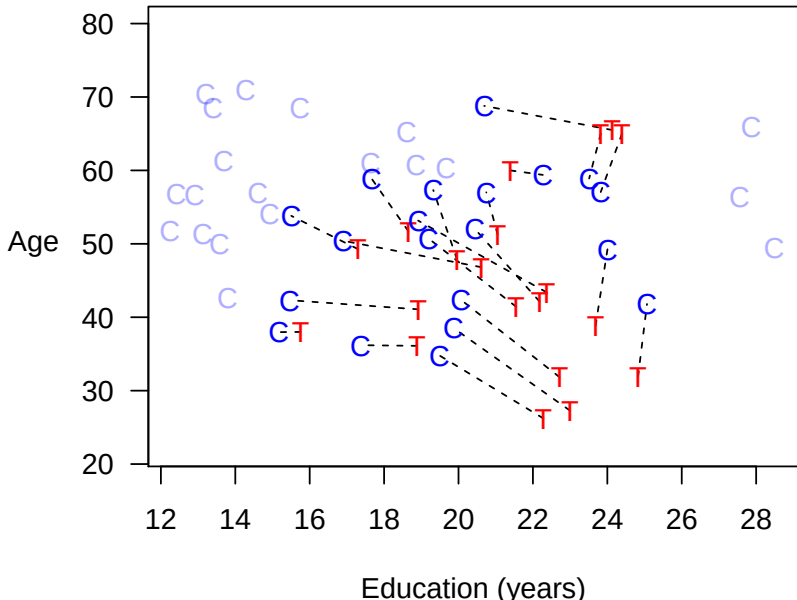
Mahalanobis Distance Matching



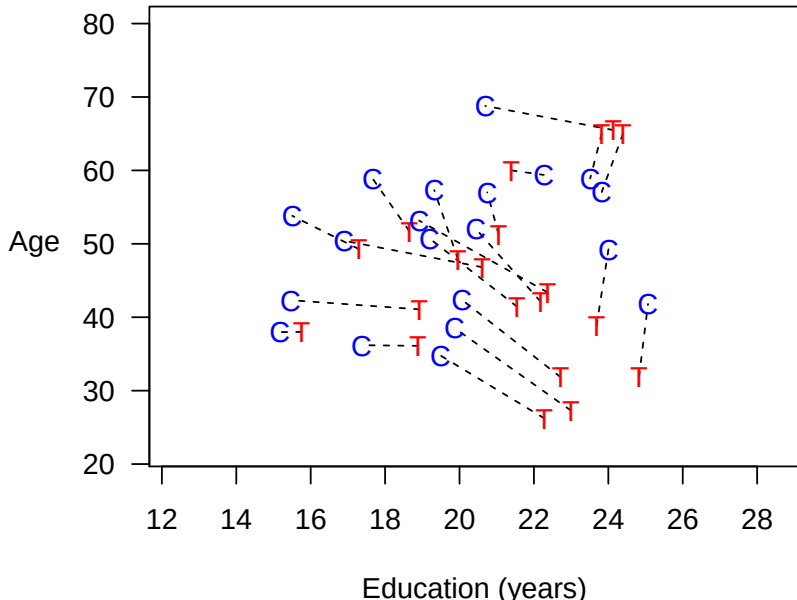
Mahalanobis Distance Matching



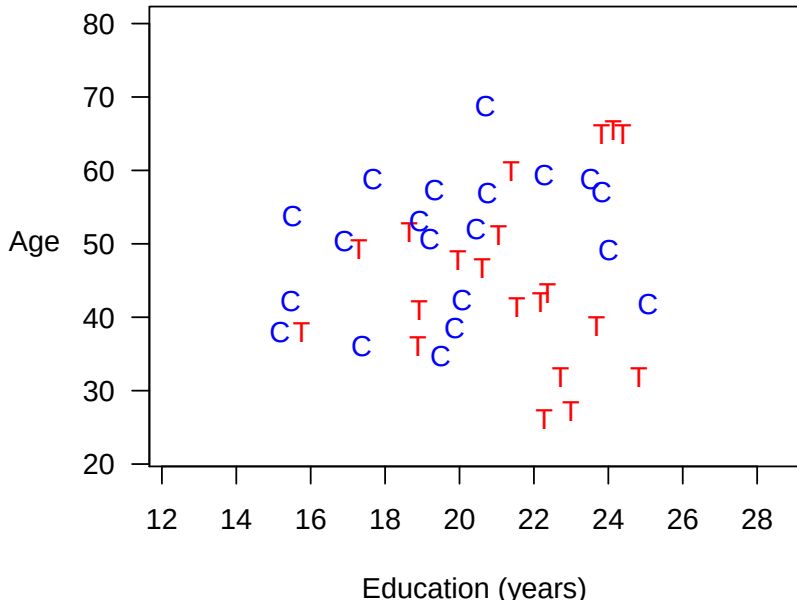
Mahalanobis Distance Matching



Mahalanobis Distance Matching

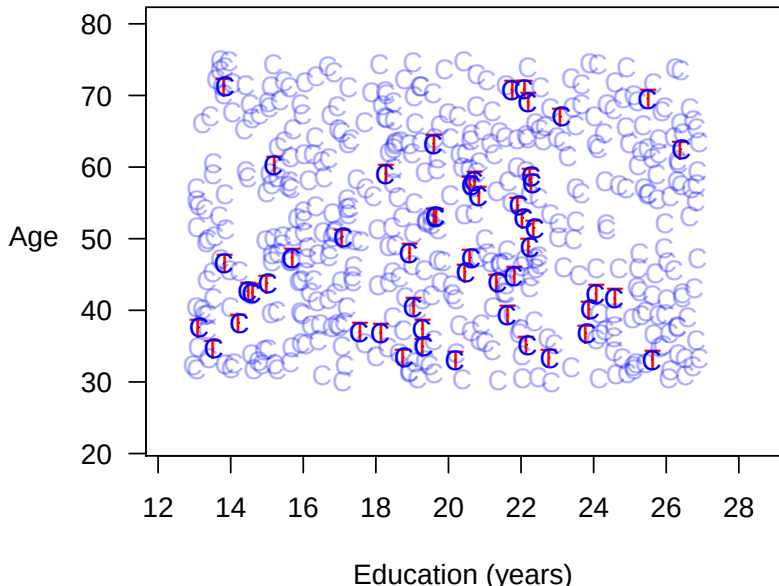


Mahalanobis Distance Matching

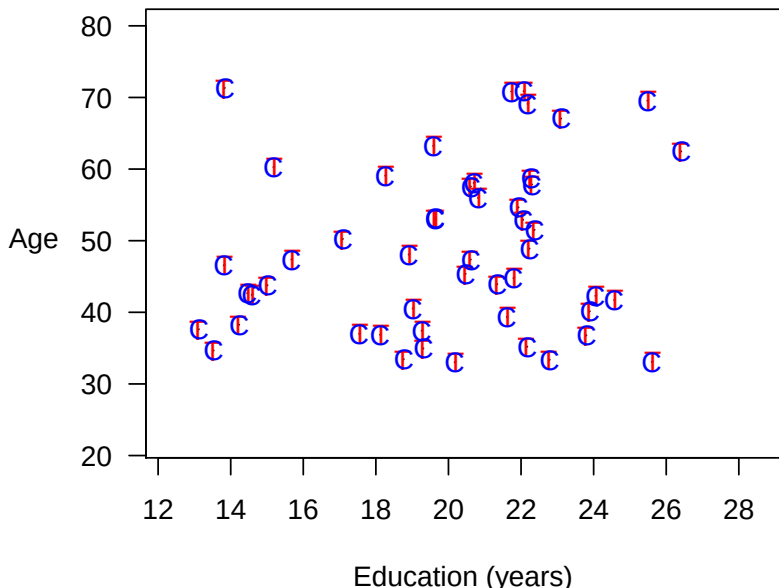


Best Case: Mahalanobis Distance Matching

Best Case: Mahalanobis Distance Matching



Best Case: Mahalanobis Distance Matching



Method 2: Coarsened Exact Matching

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. **Preprocess** (Matching)
2. **Estimation** Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)
- Apply exact matching to the coarsened X , $C(X)$

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)
- Apply exact matching to the coarsened X , $C(X)$
 - Sort observations into strata, each with unique values of $C(X)$

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)
- Apply exact matching to the coarsened X , $C(X)$
 - Sort observations into strata, each with unique values of $C(X)$
 - Prune any stratum with 0 treated or 0 control units

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)
- Apply exact matching to the coarsened X , $C(X)$
 - Sort observations into strata, each with unique values of $C(X)$
 - Prune any stratum with 0 treated or 0 control units
- Pass on original (uncoarsened) units except those pruned

2. Estimation Difference in means or a model

Method 2: Coarsened Exact Matching

(Approximates Fully Blocked Experiment)

1. Preprocess (Matching)

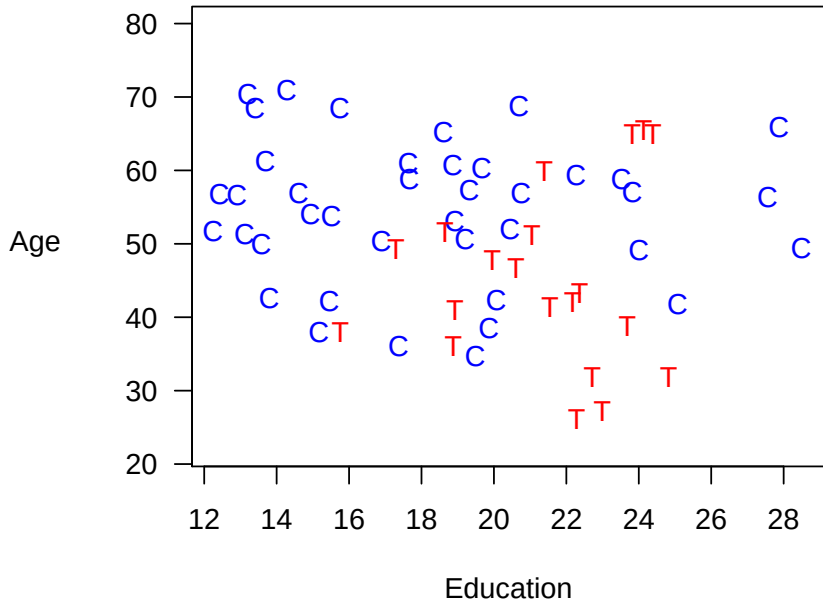
- Temporarily coarsen X as much as you're willing
 - e.g., Education (grade school, high school, college, graduate)
- Apply exact matching to the coarsened X , $C(X)$
 - Sort observations into strata, each with unique values of $C(X)$
 - Prune any stratum with 0 treated or 0 control units
- Pass on original (uncoarsened) units except those pruned

2. Estimation Difference in means or a model

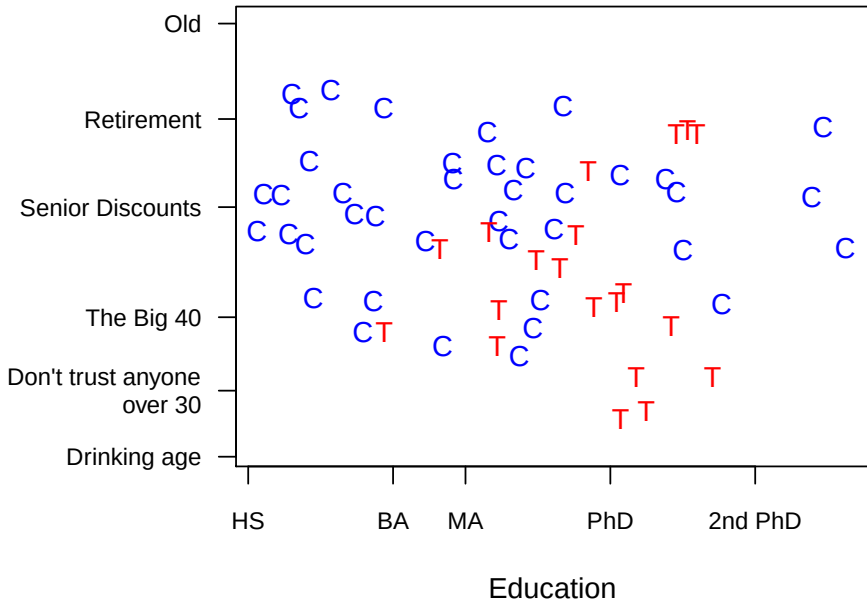
- Weight controls in each stratum to equal treateds

Coarsened Exact Matching

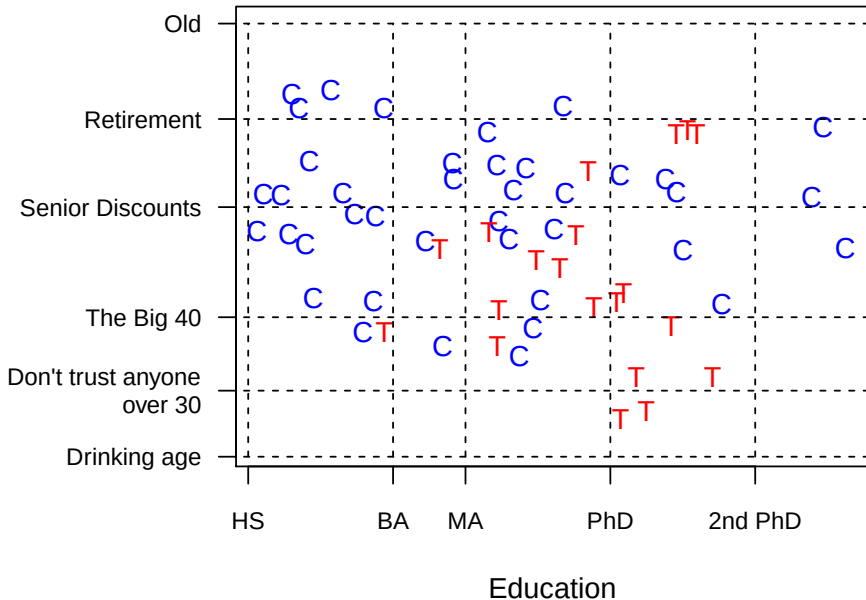
Coarsened Exact Matching



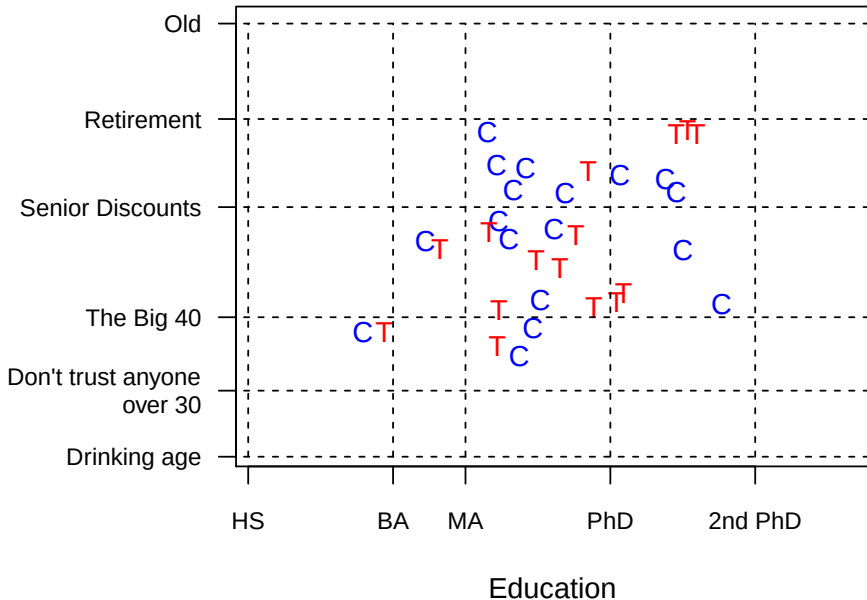
Coarsened Exact Matching



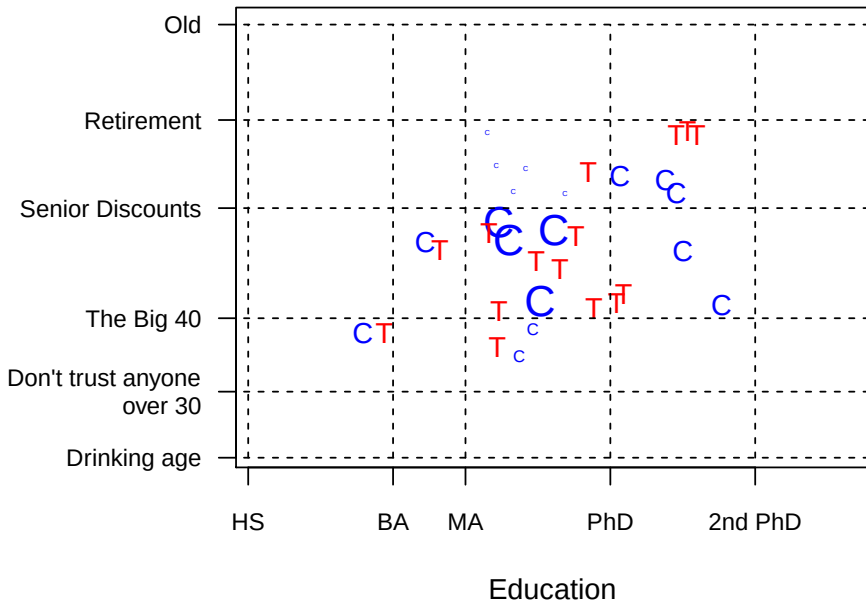
Coarsened Exact Matching



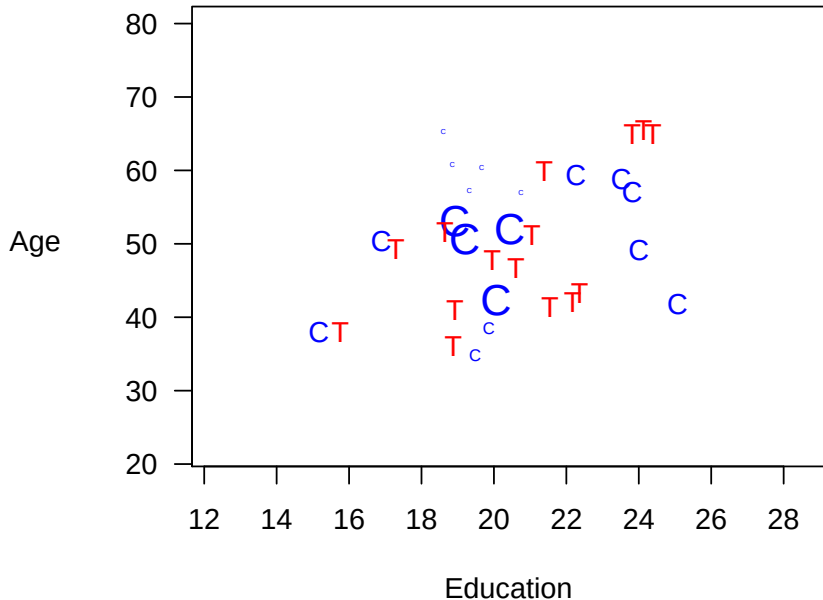
Coarsened Exact Matching



Coarsened Exact Matching

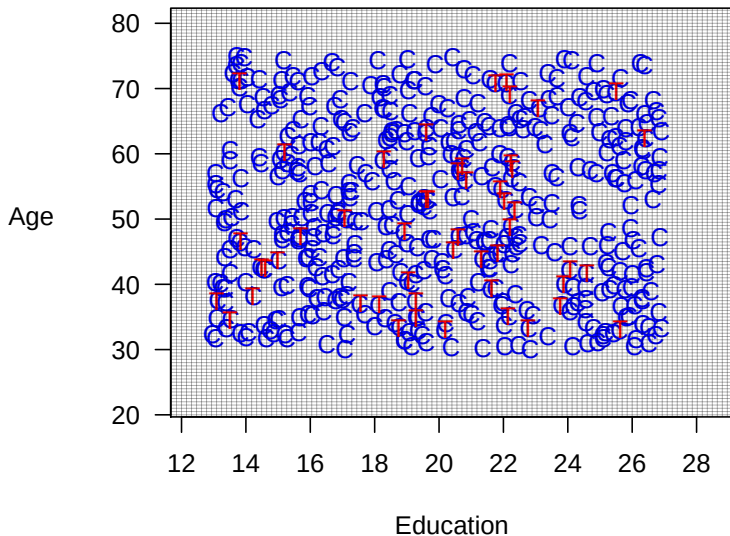


Coarsened Exact Matching

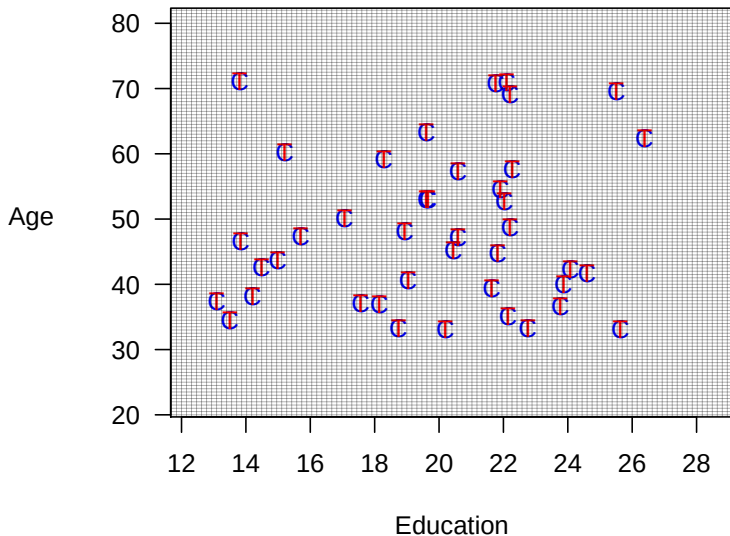


Best Case: Coarsened Exact Matching

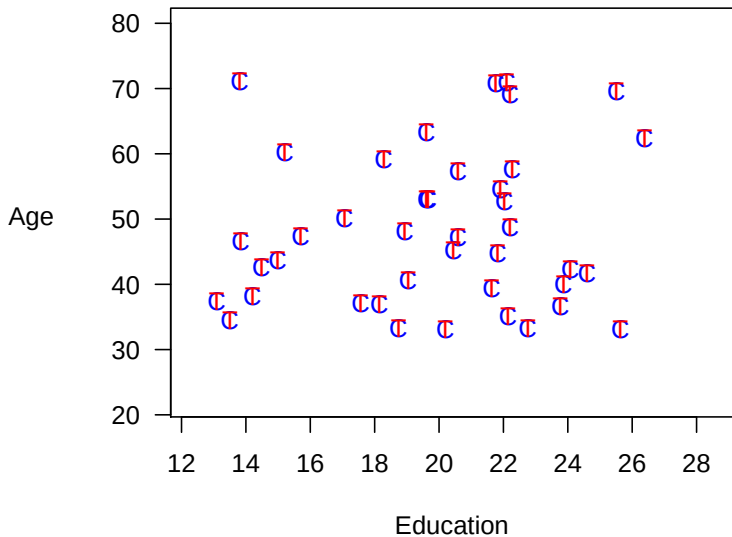
Best Case: Coarsened Exact Matching



Best Case: Coarsened Exact Matching



Best Case: Coarsened Exact Matching



Method 3: Propensity Score Matching

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. **Preprocess** (Matching)
2. **Estimation** Difference in means or a model

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

- Reduce k elements of X to scalar

$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1+e^{-X_i\beta}}$$

2. Estimation Difference in means or a model

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

- Reduce k elements of X to scalar
$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1 + e^{-X_i\beta}}$$
- Distance(X_c, X_t) = $|\pi_c - \pi_t|$

2. Estimation Difference in means or a model

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

- Reduce k elements of X to scalar
$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1 + e^{-X_i\beta}}$$
- $\text{Distance}(X_c, X_t) = |\pi_c - \pi_t|$
- Match each treated unit to the nearest control unit

2. Estimation Difference in means or a model

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

- Reduce k elements of X to scalar
$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1 + e^{-X_i\beta}}$$
- Distance(X_c, X_t) = $|\pi_c - \pi_t|$
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused

2. Estimation Difference in means or a model

Method 3: Propensity Score Matching

(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

- Reduce k elements of X to scalar
$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1 + e^{-X_i\beta}}$$
- $\text{Distance}(X_c, X_t) = |\pi_c - \pi_t|$
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused
- Prune matches if $\text{Distance} > \text{caliper}$

2. Estimation Difference in means or a model

Method 3: Propensity Score Matching

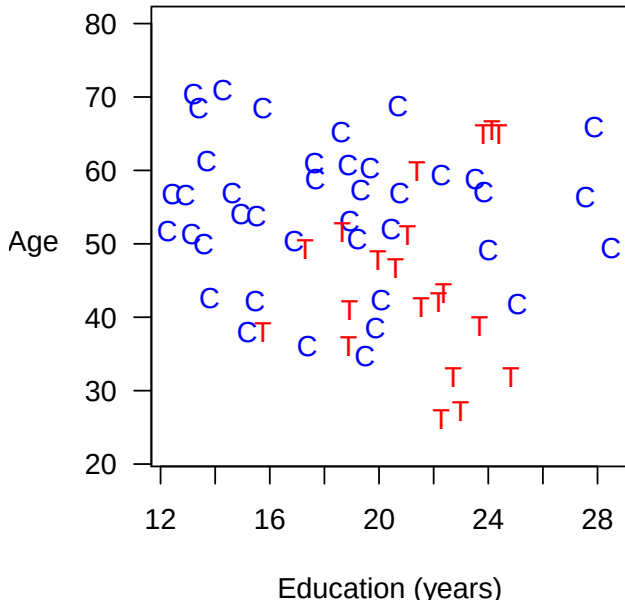
(Approximates Completely Randomized Experiment)

1. Preprocess (Matching)

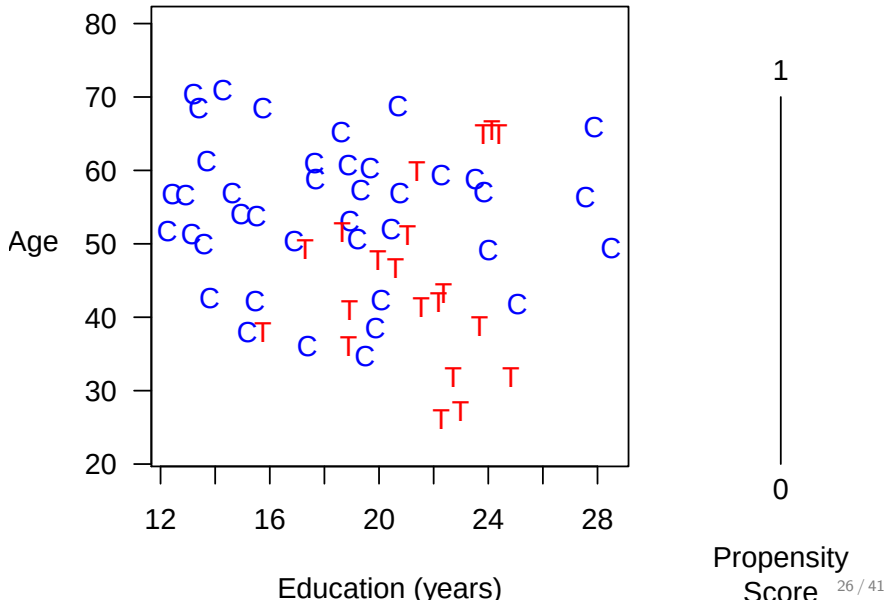
- Reduce k elements of X to scalar
$$\pi_i \equiv \Pr(T_i = 1|X) = \frac{1}{1 + e^{-X_i\beta}}$$
- $\text{Distance}(X_c, X_t) = |\pi_c - \pi_t|$
- Match each treated unit to the nearest control unit
- Control units: not reused; pruned if unused
- Prune matches if $\text{Distance} > \text{caliper}$
- (Many adjustments available to this basic method)

2. Estimation Difference in means or a model

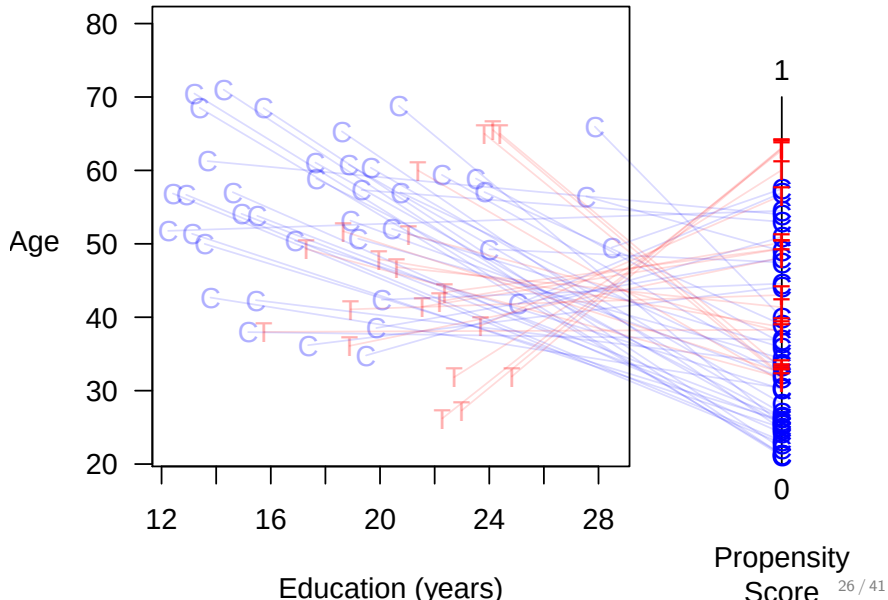
Propensity Score Matching



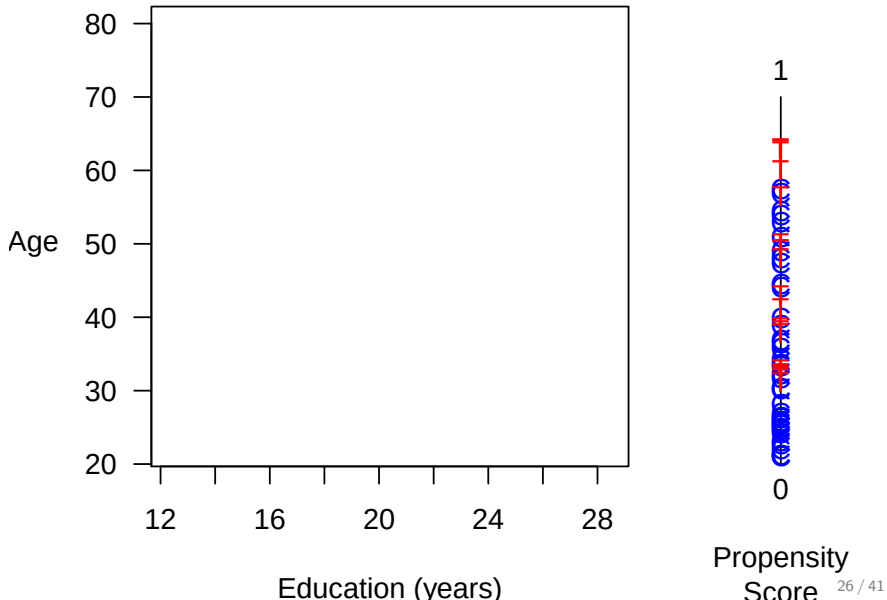
Propensity Score Matching



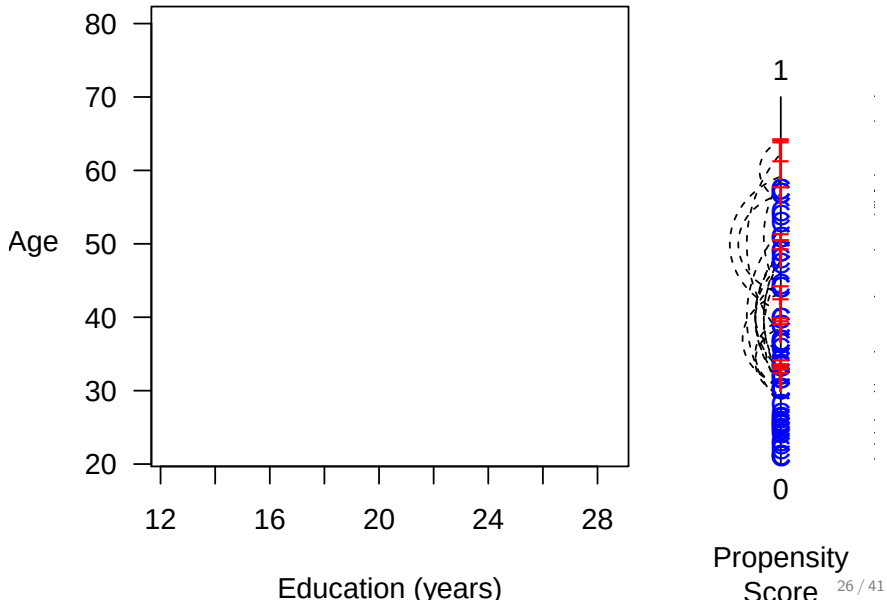
Propensity Score Matching



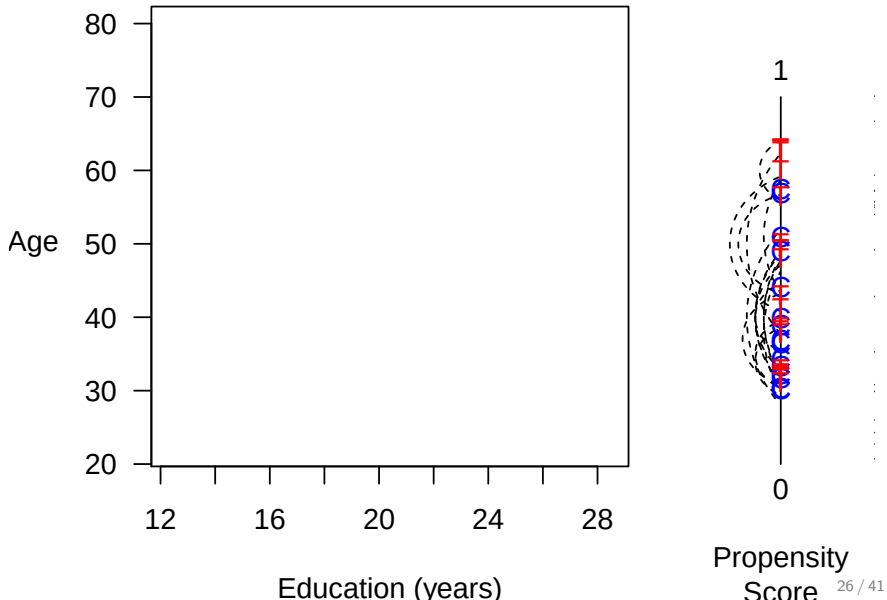
Propensity Score Matching



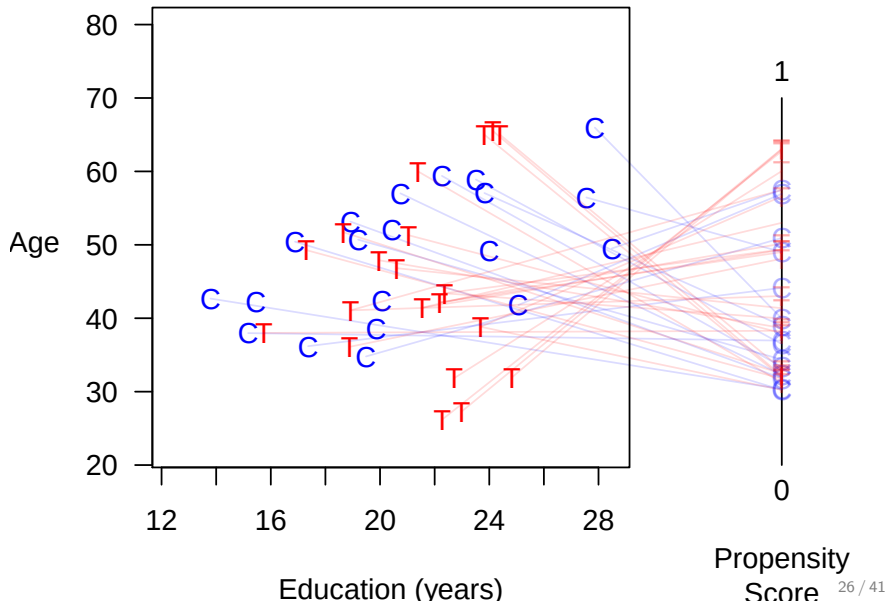
Propensity Score Matching



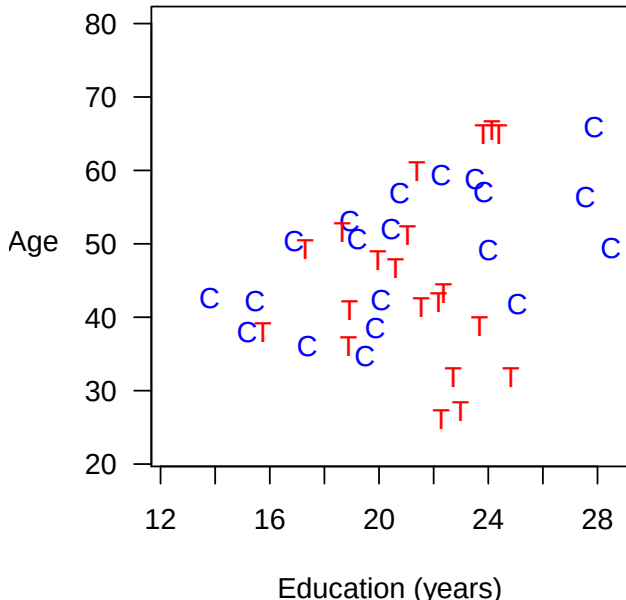
Propensity Score Matching



Propensity Score Matching

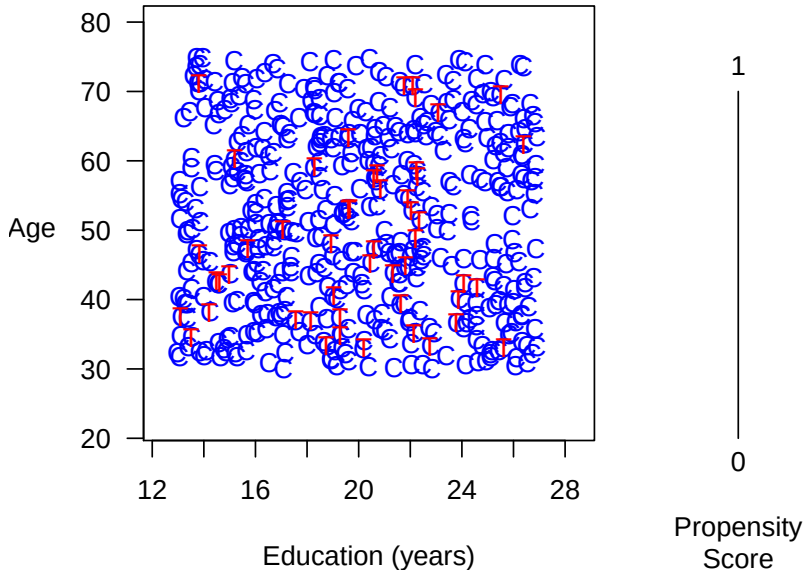


Propensity Score Matching

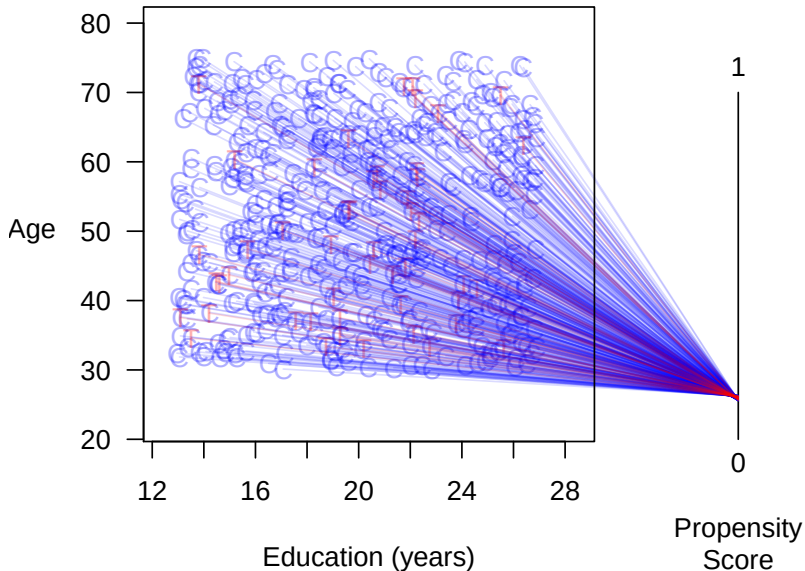


Best Case: Propensity Score Matching

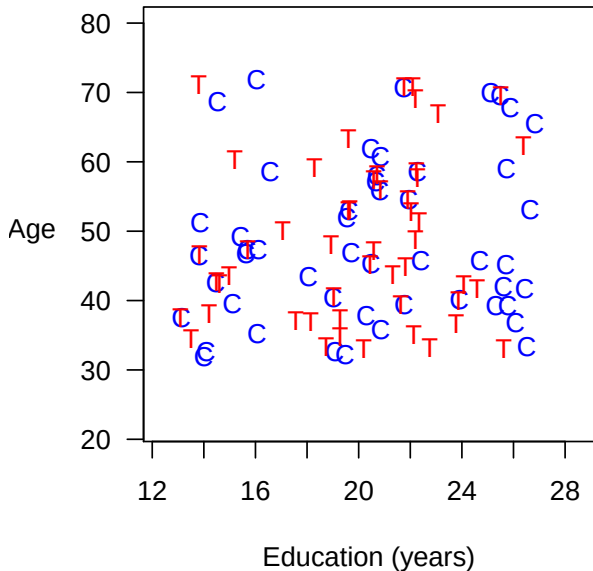
Best Case: Propensity Score Matching



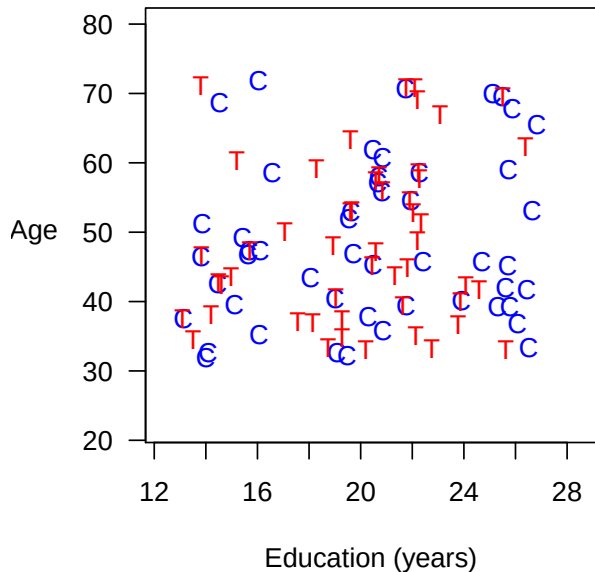
Best Case: Propensity Score Matching



Best Case: Propensity Score Matching



Best Case: Propensity Score Matching is Suboptimal



Random Pruning Increases Imbalance

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- Continuous example

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- Continuous example
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- Discrete example
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- Continuous example
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units
 - Measure of imbalance: squared difference in means d^2 , where $d = \bar{X}_t - \bar{X}_c$

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- **Discrete example**
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- **Continuous example**
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units
 - Measure of imbalance: squared difference in means d^2 , where $d = \bar{X}_t - \bar{X}_c$
 - $E(d^2) = V(d) \propto 1/n$ (note: $E(d) = 0$)

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- **Discrete example**
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
 - 2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
 - 2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- **Continuous example**
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units
 - Measure of imbalance: squared difference in means d^2 , where $d = \bar{X}_t - \bar{X}_c$
 - $E(d^2) = V(d) \propto 1/n$ (note: $E(d) = 0$)
 - Random pruning $\rightsquigarrow n$ declines $\rightsquigarrow E(d^2)$ increases

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- **Discrete example**
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- **Continuous example**
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units
 - Measure of imbalance: squared difference in means d^2 , where $d = \bar{X}_t - \bar{X}_c$
 - $E(d^2) = V(d) \propto 1/n$ (note: $E(d) = 0$)
 - Random pruning $\rightsquigarrow n$ declines $\rightsquigarrow E(d^2)$ increases
 - $\implies$ random pruning increases imbalance

Random Pruning Increases Imbalance

Deleting data only helps if you're careful!

- “Random pruning”: pruning process is independent of X
- **Discrete example**
 - Sex-balanced dataset: treateds M_t, F_t , controls M_c, F_c
 - Randomly prune 1 treated & 1 control $\rightsquigarrow$ 4 possible datasets:
2 balanced $\{M_t, M_c\}, \{F_t, F_c\}$
2 imbalanced $\{M_t, F_c\}, \{F_t, M_c\}$
 - $\implies$ random pruning increases imbalance
- **Continuous example**
 - Dataset: $T \in \{0, 1\}$ randomly assigned; X any fixed variable; with n units
 - Measure of imbalance: squared difference in means d^2 , where $d = \bar{X}_t - \bar{X}_c$
 - $E(d^2) = V(d) \propto 1/n$ (note: $E(d) = 0$)
 - Random pruning $\rightsquigarrow n$ declines $\rightsquigarrow E(d^2)$ increases
 - $\implies$ random pruning increases imbalance
- **Result is completely general (see math in the paper)**

PSM's Statistical Properties

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes
 - *Efficient* relative to complete randomization, but

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes
 - *Efficient* relative to complete randomization, but
 - *Inefficient* relative to (the more powerful) full blocking

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes
 - *Efficient* relative to complete randomization, but
 - *Inefficient* relative to (the more powerful) full blocking
 - Other methods dominate:

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t$$

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes
 - *Efficient* relative to complete randomization, but
 - *Inefficient* relative to (the more powerful) full blocking
 - Other methods dominate:
 $X_c = X_t \implies \pi_c = \pi_t$ but
 $\pi_c = \pi_t \not\Rightarrow X_c = X_t$
2. The PSM Paradox: When you do “better,” you do worse

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes
 - *Efficient* relative to complete randomization, but
 - *Inefficient* relative to (the more powerful) full blocking
 - Other methods dominate:
 $X_c = X_t \implies \pi_c = \pi_t$ but
 $\pi_c = \pi_t \not\Rightarrow X_c = X_t$
2. The PSM Paradox: When you do “better,” you do worse
 - Background: Random matching increases imbalance

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning)

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata)

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\implies X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\implies X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\implies X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence $\rightsquigarrow$ Bias

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence $\rightsquigarrow$ Bias
- If the data have no good matches, the paradox won't be a problem but you're cooked anyway.

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence $\rightsquigarrow$ Bias
- If the data have no good matches, the paradox won't be a problem but you're cooked anyway.
- Doesn't PSM solve the curse of dimensionality problem?

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence $\rightsquigarrow$ Bias
- If the data have no good matches, the paradox won't be a problem but you're cooked anyway.
- Doesn't PSM solve the curse of dimensionality problem? Nope.

PSM's Statistical Properties

1. Low Standards: Sometimes helps, never optimizes

- *Efficient* relative to complete randomization, but
- *Inefficient* relative to (the more powerful) full blocking
- Other methods dominate:

$$X_c = X_t \implies \pi_c = \pi_t \text{ but}$$

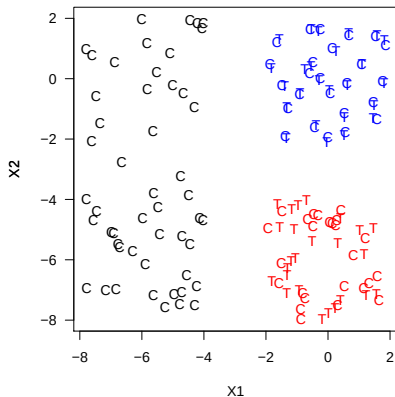
$$\pi_c = \pi_t \not\Rightarrow X_c = X_t$$

2. The PSM Paradox: When you do “better,” you do worse

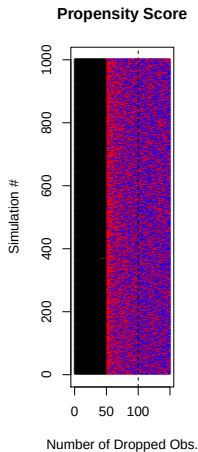
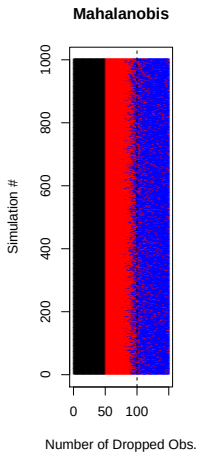
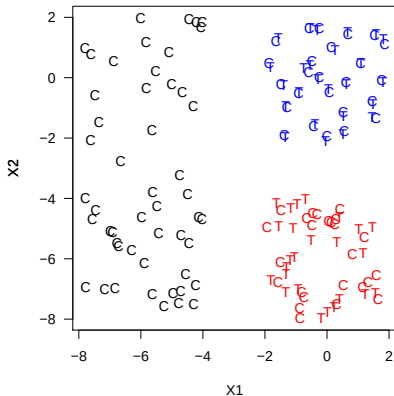
- Background: Random matching increases imbalance
- When PSM approximates complete randomization (to begin with or, after some pruning) $\rightsquigarrow$ all $\hat{\pi} \approx 0.5$ (or constant within strata) $\rightsquigarrow$ pruning at random $\rightsquigarrow$ Imbalance $\rightsquigarrow$ Inefficiency $\rightsquigarrow$ Model dependence $\rightsquigarrow$ Bias
- If the data have no good matches, the paradox won't be a problem but you're cooked anyway.
- Doesn't PSM solve the curse of dimensionality problem? Nope. The PSM Paradox gets worse with more covariates

PSM is Blind Where Other Methods Can See

PSM is Blind Where Other Methods Can See

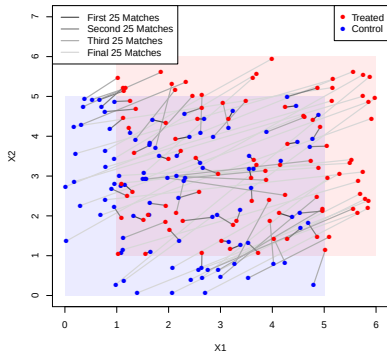


PSM is Blind Where Other Methods Can See

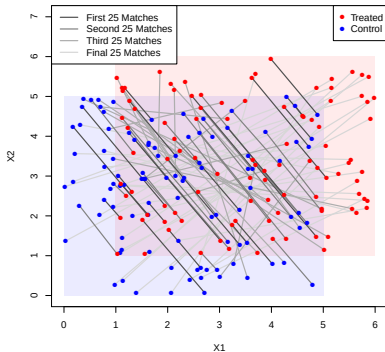


What Does PSM Match?

MDM Matches



PSM Matches

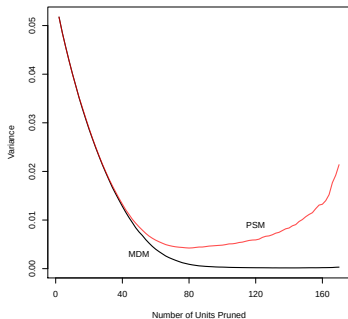


Controls: $X_1, X_2 \sim \text{Uniform}(0,5)$

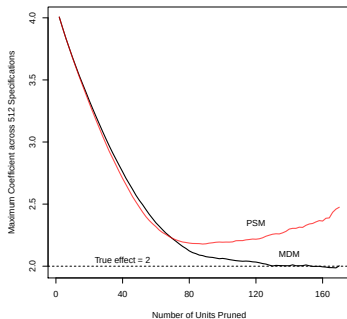
Treateds: $X_1, X_2 \sim \text{Uniform}(1,6)$

PSM Increases Model Dependence & Bias

Model Dependence



Bias

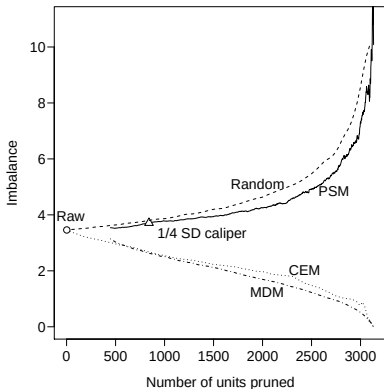


$$Y_i = 2T_i + X_{1i} + X_{2i} + \epsilon_i$$
$$\epsilon_i \sim N(0, 1)$$

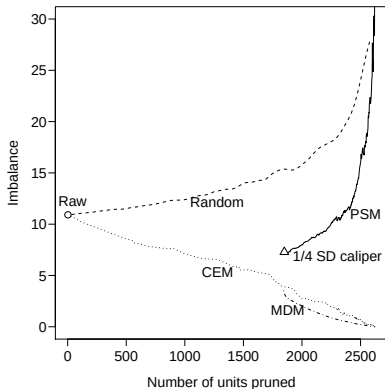
The Propensity Score Paradox in Real Data

The Propensity Score Paradox in Real Data

Finkel et al. (JOP, 2012)

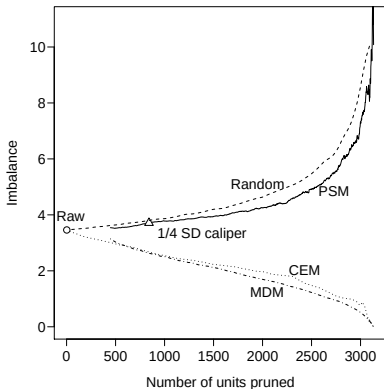


Nielsen et al. (AJPS, 2011)

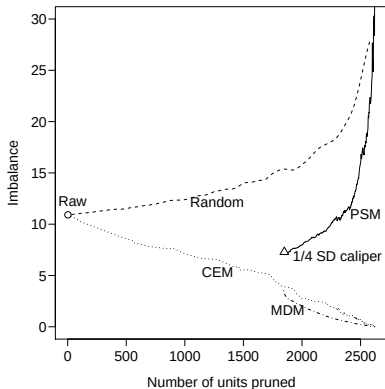


The Propensity Score Paradox in Real Data

Finkel et al. (JOP, 2012)



Nielsen et al. (AJPS, 2011)



Similar pattern for > 20 other real data sets we checked

The Matching Frontier

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use
- No need to choose or use a matching method

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use
- No need to choose or use a matching method
- All solutions are optimal

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use
- No need to choose or use a matching method
- All solutions are optimal
- No iteration or diagnostics required

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use
- No need to choose or use a matching method
- All solutions are optimal
- No iteration or diagnostics required
- No cherry picking possible; you see everything optimal

The Matching Frontier

- **Frontier** = matched dataset with lowest imbalance for each n
- Bias-Variance trade off $\rightsquigarrow$ Imbalance- n Trade Off
- Simple to use
- No need to choose or use a matching method
- All solutions are optimal
- No iteration or diagnostics required
- No cherry picking possible; you see everything optimal
- Choose an imbalance metric, then run.

How hard is the frontier to calculate?

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast
 - operate as “greedy” but we prove are optimal

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast
 - operate as “greedy” but we prove are optimal
 - do not require evaluating every subset

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast
 - operate as “greedy” but we prove are optimal
 - do not require evaluating every subset
 - work with very large data sets

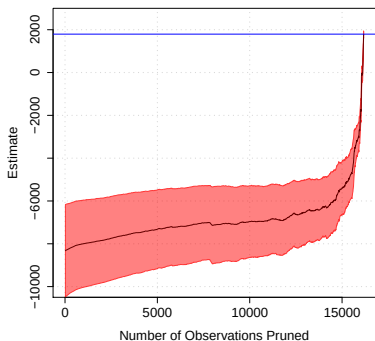
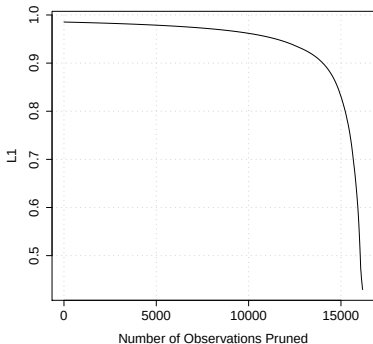
How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast
 - operate as “greedy” but we prove are optimal
 - do not require evaluating every subset
 - work with very large data sets
 - is the exact frontier (no approximation or estimation)

How hard is the frontier to calculate?

- Consider 1 point on the SATT frontier:
 - Start with matrix of N control units X_0
 - Calculate imbalance for all $\binom{N}{n}$ subsets of rows of X_0
 - Choose subset with lowest imbalance
- Evaluations needed to compute the entire frontier:
 - $\binom{N}{n}$ evaluations for each sample size $n = N, N - 1, \dots, 1$
 - The combination is the (gargantuan) “power set”
 - e.g., $N > 300$ requires more imbalance evaluations than elementary particles in the universe
 - $\rightsquigarrow$ It's **hard** to calculate!
- We develop algorithms for the (optimal) frontier which:
 - runs very fast
 - operate as “greedy” but we prove are optimal
 - do not require evaluating every subset
 - work with very large data sets
 - is the exact frontier (no approximation or estimation)
 - $\rightsquigarrow$ It's **easy** to calculate!

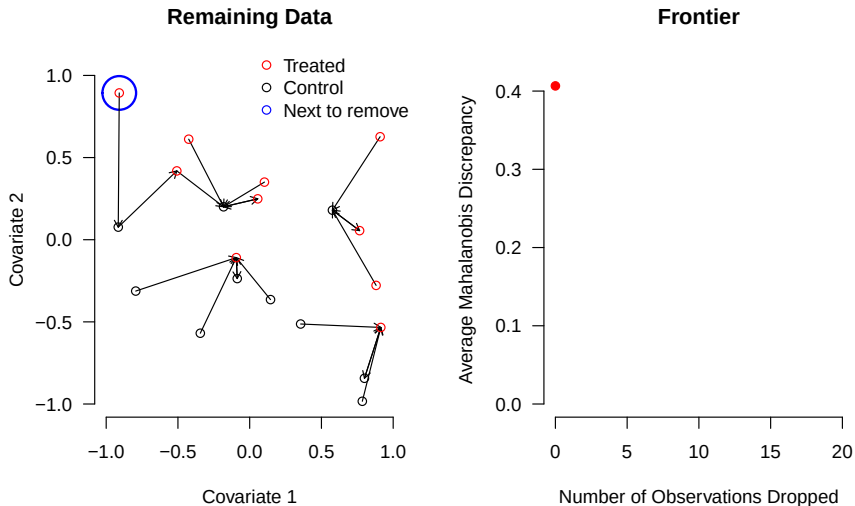
Job Training Data: Frontier and Causal Estimates



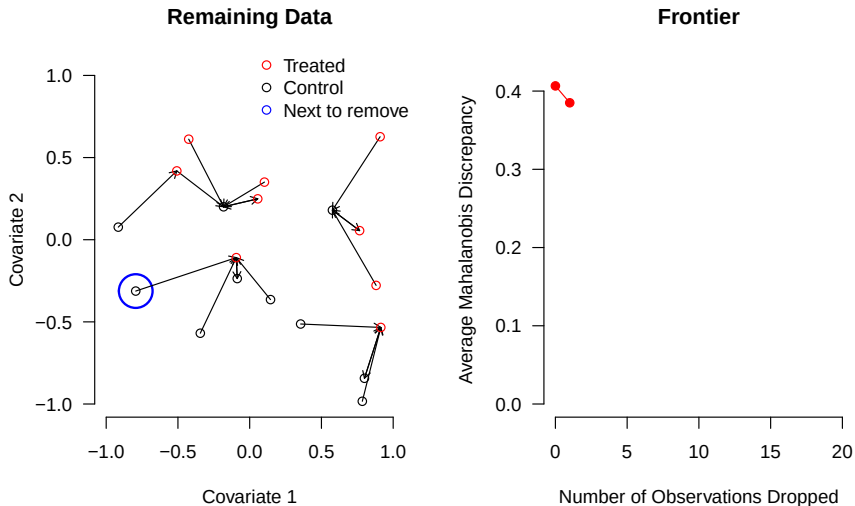
- 185 Ts; pruning most 16,252 Cs won't increase variance much
- Huge bias-variance trade-off after pruning most Cs
- Estimates converge to experiment after removing bias
- No mysteries: basis of inference clearly revealed

Constructing the FSATT Mahalanobis Frontier

Constructing the FSATT Mahalanobis Frontier

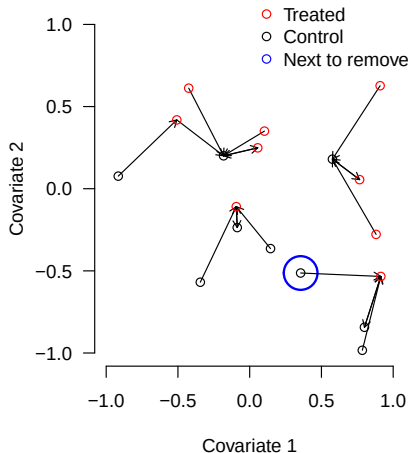


Constructing the FSATT Mahalanobis Frontier

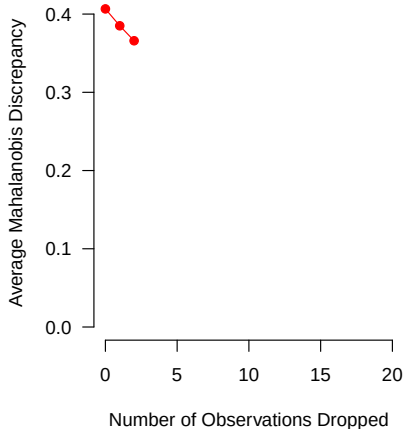


Constructing the FSATT Mahalanobis Frontier

Remaining Data

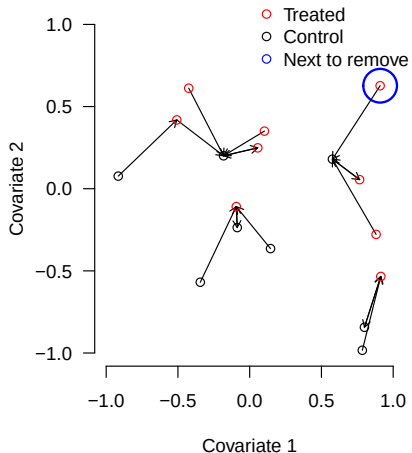


Frontier

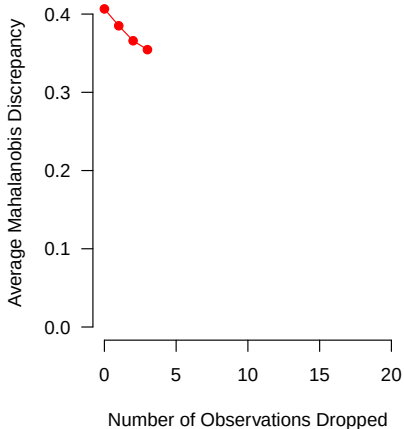


Constructing the FSATT Mahalanobis Frontier

Remaining Data

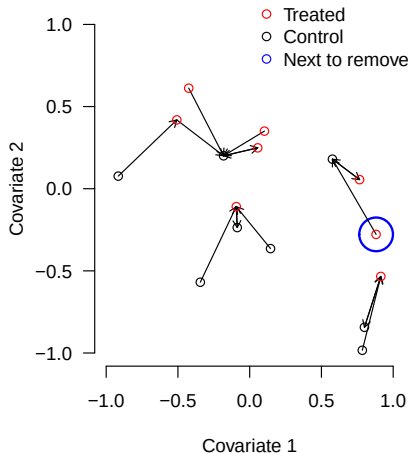


Frontier

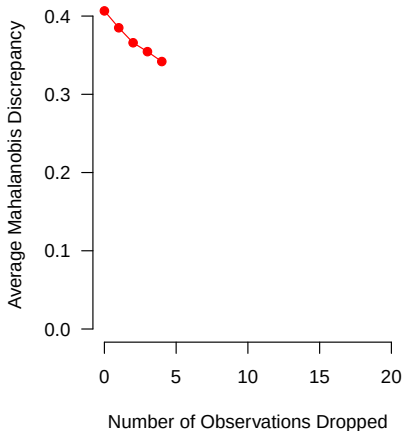


Constructing the FSATT Mahalanobis Frontier

Remaining Data

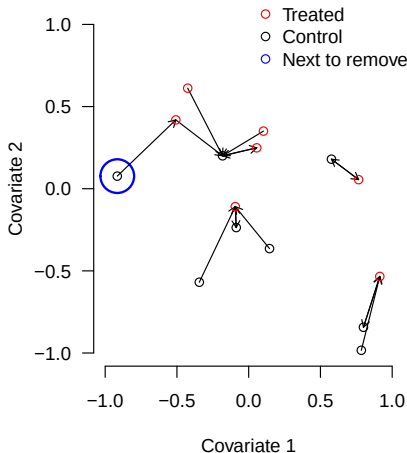


Frontier

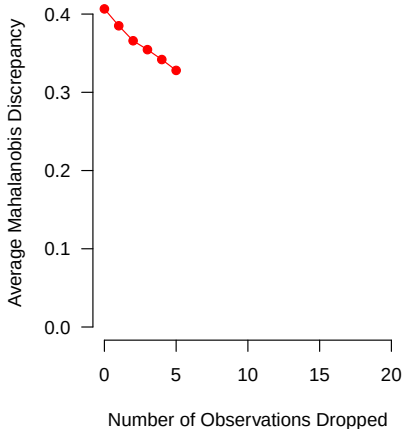


Constructing the FSATT Mahalanobis Frontier

Remaining Data

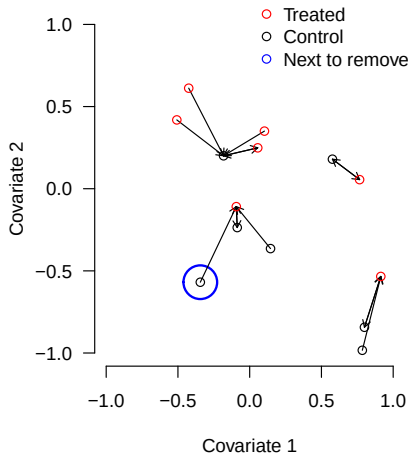


Frontier

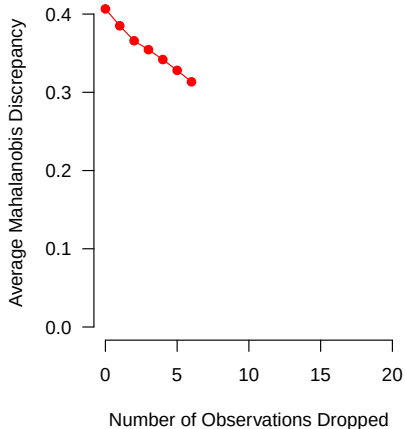


Constructing the FSATT Mahalanobis Frontier

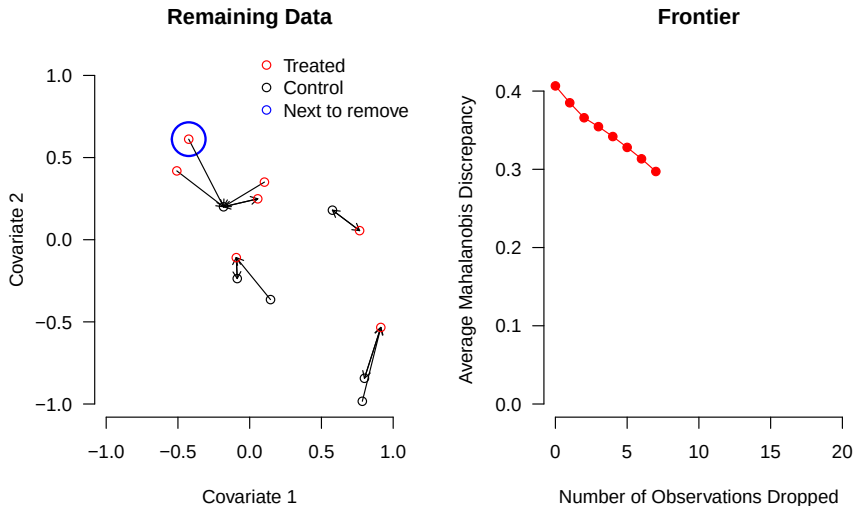
Remaining Data



Frontier

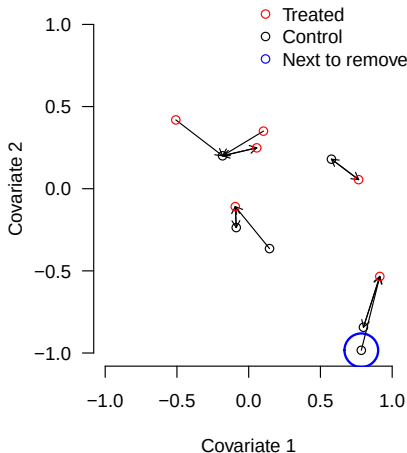


Constructing the FSATT Mahalanobis Frontier

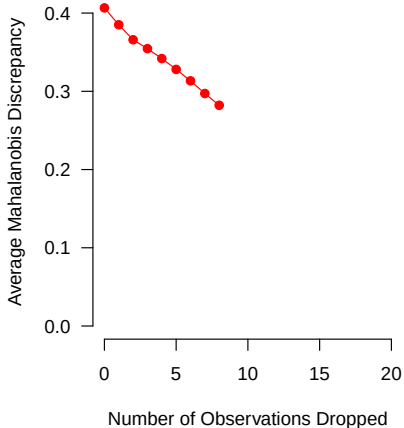


Constructing the FSATT Mahalanobis Frontier

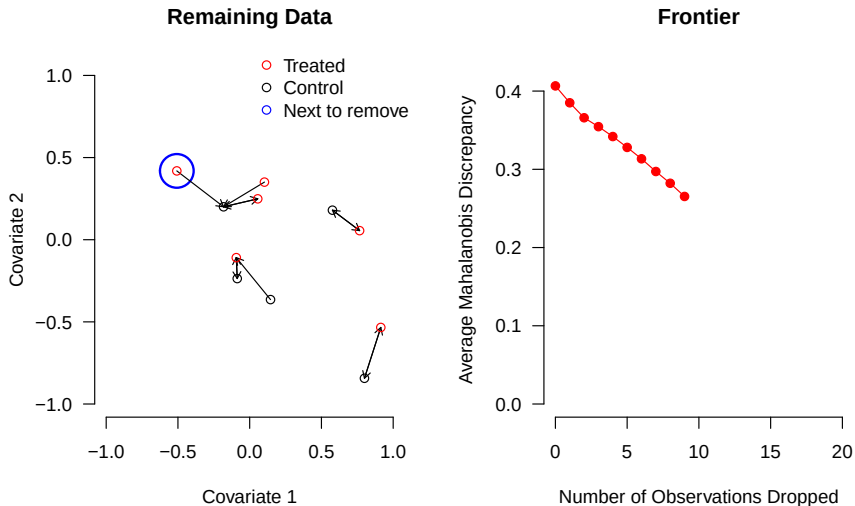
Remaining Data



Frontier

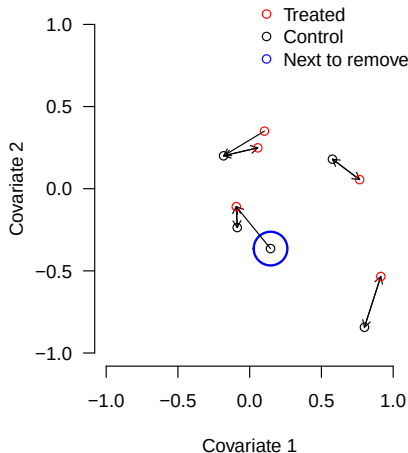


Constructing the FSATT Mahalanobis Frontier

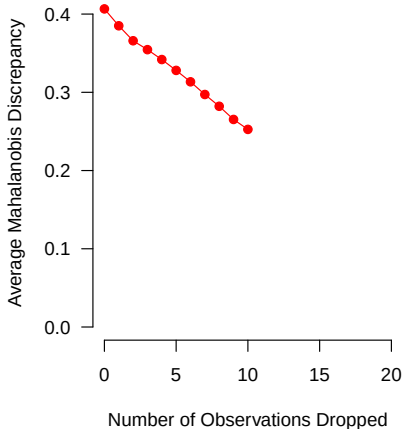


Constructing the FSATT Mahalanobis Frontier

Remaining Data

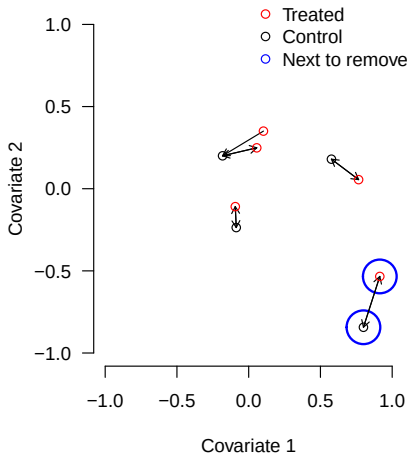


Frontier

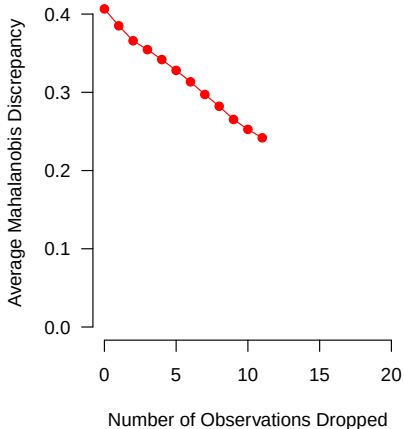


Constructing the FSATT Mahalanobis Frontier

Remaining Data

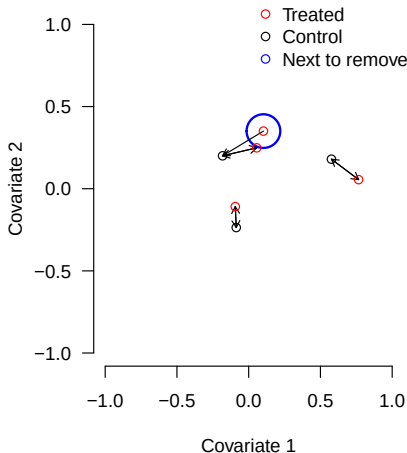


Frontier

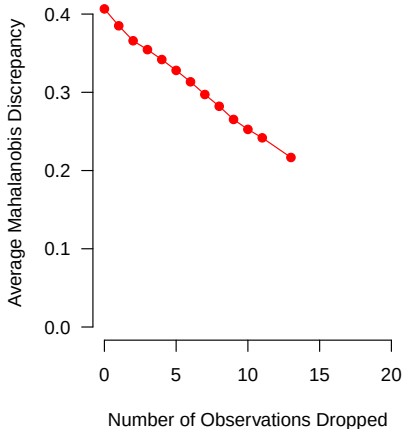


Constructing the FSATT Mahalanobis Frontier

Remaining Data

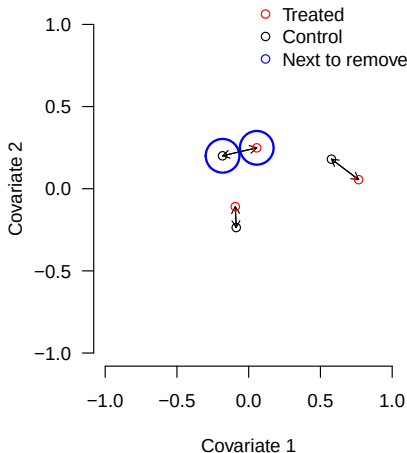


Frontier

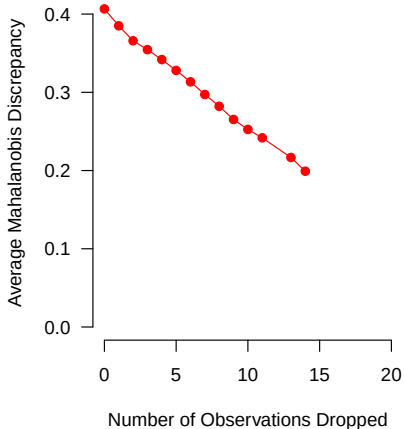


Constructing the FSATT Mahalanobis Frontier

Remaining Data

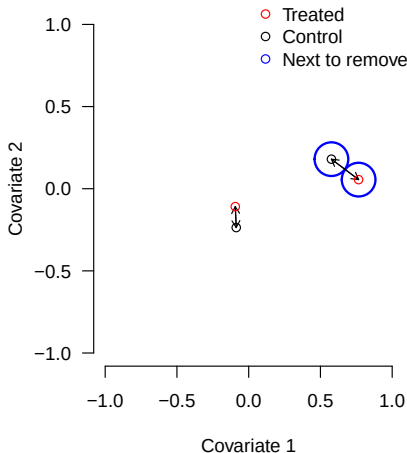


Frontier

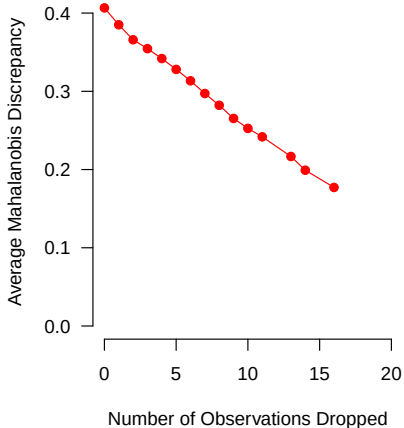


Constructing the FSATT Mahalanobis Frontier

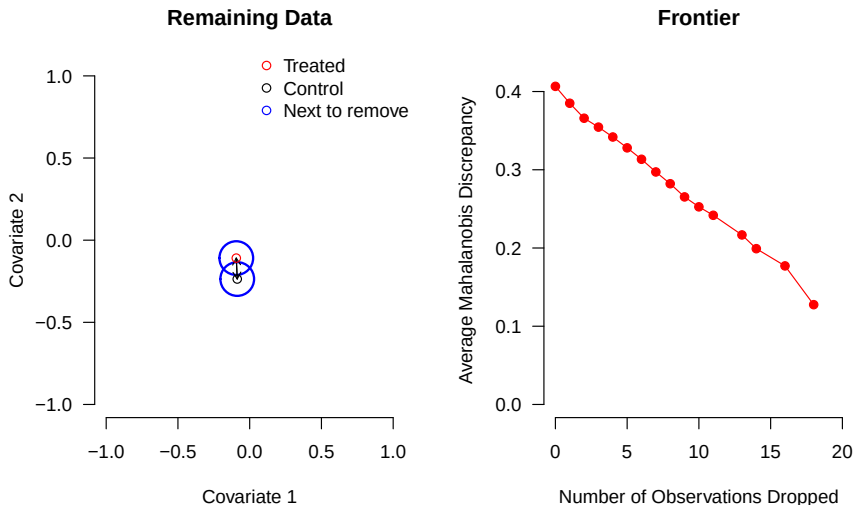
Remaining Data



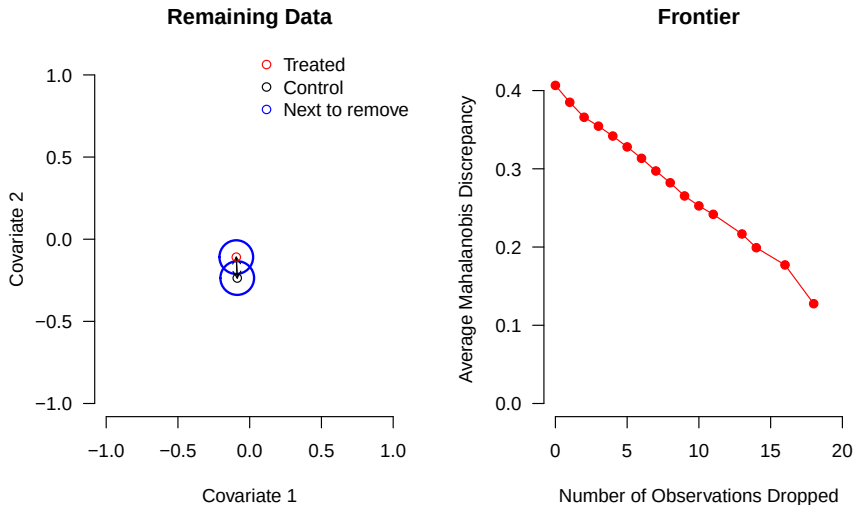
Frontier



Constructing the FSATT Mahalanobis Frontier

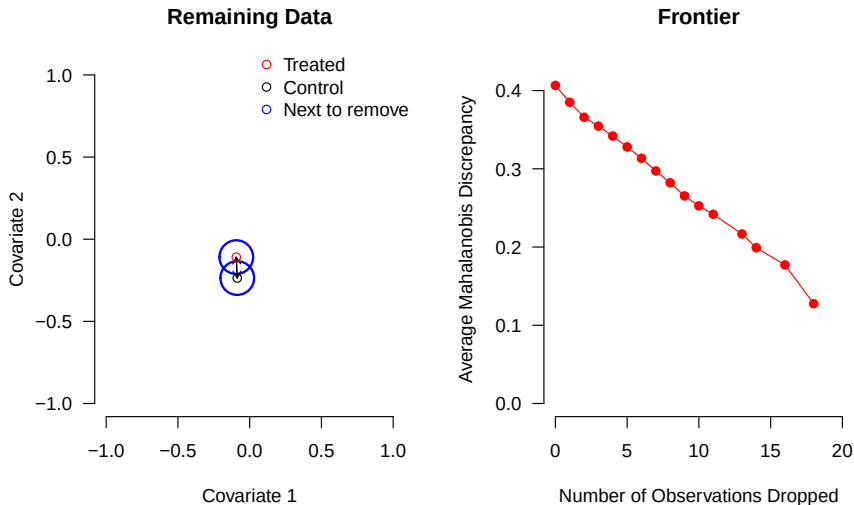


Constructing the FSATT Mahalanobis Frontier



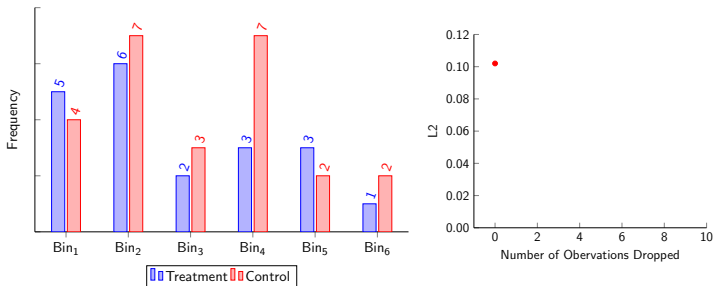
- Warning: figure omits details and the proof!

Constructing the FSATT Mahalanobis Frontier

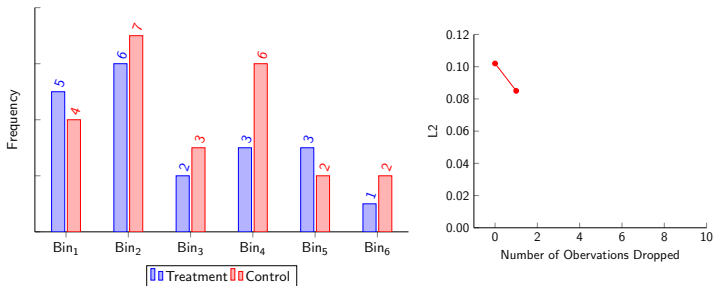


- Warning: figure omits details and the proof!
- Very fast; works with any continuous imbalance metric

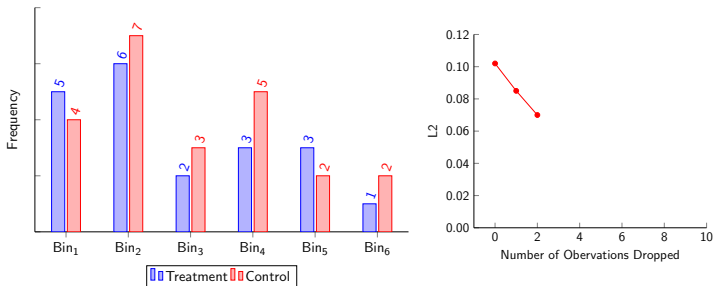
Constructing the L1/L2 SATT Frontier



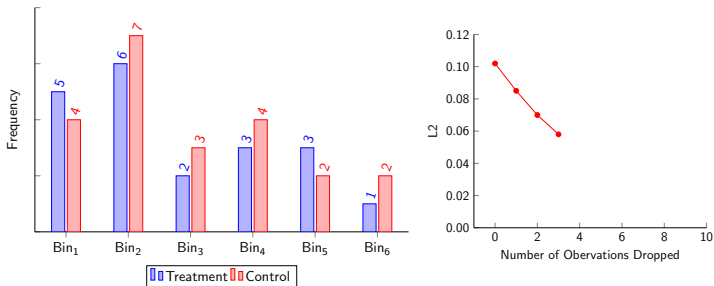
Constructing the L1/L2 SATT Frontier



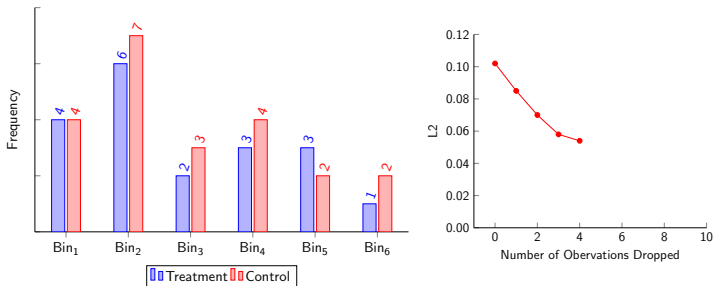
Constructing the L1/L2 SATT Frontier



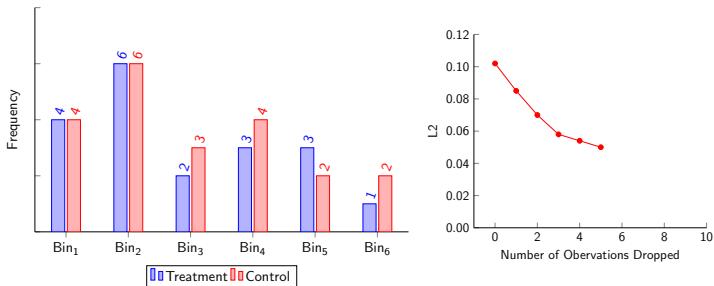
Constructing the L1/L2 SATT Frontier



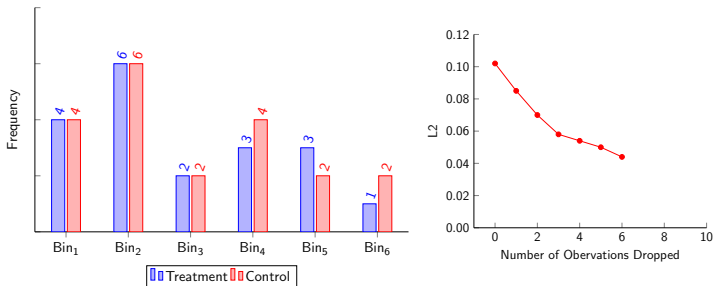
Constructing the L1/L2 SATT Frontier



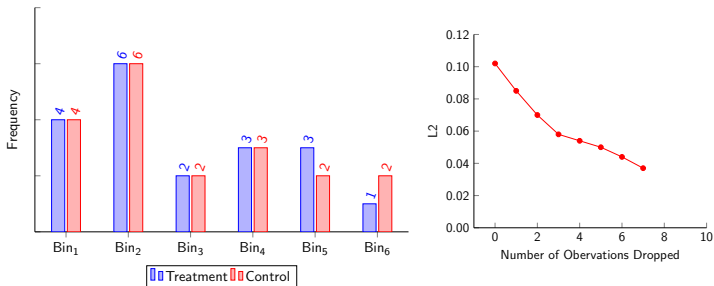
Constructing the L1/L2 SATT Frontier



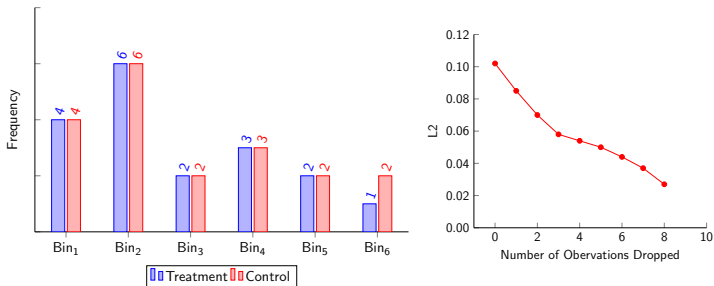
Constructing the L1/L2 SATT Frontier



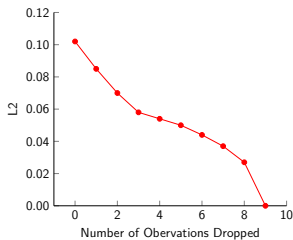
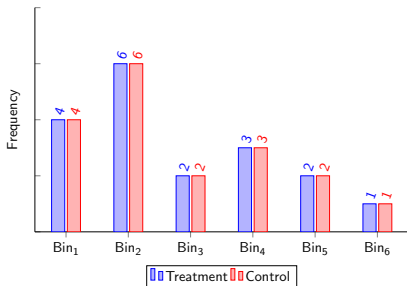
Constructing the L1/L2 SATT Frontier



Constructing the L1/L2 SATT Frontier



Constructing the L1/L2 SATT Frontier



Constructing the L1/L2 SATT Frontier

Constructing the L1/L2 SATT Frontier

- Warning: This figure omits some technical details too!

Constructing the L1/L2 SATT Frontier

- Warning: This figure omits some technical details too!
- Works very fast, even with very large data sets

Conclusions

Conclusions

- Propensity score matching:

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM:

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates;

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data;

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support;

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score;

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- A Simple and Powerful Method: CEM

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- A Simple and Powerful Method: CEM
- A New General Approach: The Matching Frontier

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- A Simple and Powerful Method: CEM
- A New General Approach: The Matching Frontier
 - Fast; easy; no iteration; Software: MatchingFrontier

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- A Simple and Powerful Method: CEM
- A New General Approach: The Matching Frontier
 - Fast; easy; no iteration; Software: MatchingFrontier
 - No need to choose among matching methods

Conclusions

- **Propensity score matching:**
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- **A Simple and Powerful Method:** CEM
- **A New General Approach:** The Matching Frontier
 - Fast; easy; no iteration; Software: MatchingFrontier
 - No need to choose among matching methods
 - Optimal results from your choice of imbalance metric

Conclusions

- Propensity score matching:
 - Approximates complete, not fully blocked, experiments
 - Ignores information; exacerbates model dependence
 - Some mistakes with PSM: Controlling for irrelevant covariates; Adjusting experimental data; Reestimating propensity score after eliminating noncommon support; 1/4 caliper on propensity score; Not switching to other methods.
- A Simple and Powerful Method: CEM
- A New General Approach: The Matching Frontier
 - Fast; easy; no iteration; Software: MatchingFrontier
 - No need to choose among matching methods
 - Optimal results from your choice of imbalance metric
- $\rightsquigarrow$ Using more information is simpler and more powerful

For more information, articles, & software

GaryKing.org